

DiD Notes

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Preface

Difference-in-differences (DiD) is among the most widely used identification strategies in empirical economics. For decades, the two-way fixed effects (TWFE) regression served as its canonical implementation. Beginning in the late 2010s, however, a surge of methodological work revealed that TWFE can produce severely biased — and even sign-reversed — estimates when treatment adoption is staggered and effects are heterogeneous. These notes document the resulting “new DiD” literature: its theoretical foundations, its failure modes, and the modern estimators developed in response.

Structure

- **Part I — Introduction.** The canonical 2×2 setup, the potential outcomes framework and ATT, and TWFE as a multi-period generalisation. The closing section surveys the recent methodological debates that motivate the remainder of the notes.
- **Part II — Identification Assumptions.** The core of DiD identification is the parallel trends assumption (PTA). Roth (2022) showed that the common practice of conditioning on a passed pre-trend test distorts the sampling distribution of post-treatment estimates, producing size distortions and reduced power. When the PTA is plausible only after conditioning on observables, Abadie (2005) proposed an IPW-DiD estimator, and Sant’Anna & Zhao (2020) extended this to a doubly robust (DR-DiD) procedure; for high-dimensional covariates, Chernozhukov et al. (2018) developed double/debiased machine learning (DML), adapted to the DiD setting by Chang (2020). When unobserved time-varying shocks threaten the PTA entirely, Arkhangelsky et al. (2021) combined synthetic control weights with DiD (SDID), and Rambachan & Roth (2023) proposed honest sensitivity analysis that bounds inference under partial violations of parallel trends. The part also covers partial compliance: Imbens & Angrist (1994) established the LATE framework, and Chaisemartin & d’Haultfoeuille (2017) extended it to fuzzy DiD via the Wald-DiD estimator.
- **Part III — Staggered Adoption.** Goodman-Bacon (2021) showed that under staggered treatment timing the TWFE coefficient is a variance-weighted average of all possible 2×2 sub-comparisons — some of which carry negative weights, because already-treated units are used as implicit controls for later-treated units. This diagnosis sparked a wave of robust estimators: Callaway & Sant’Anna (2021) proposed group-time ATTs using never-

or not-yet-treated units as clean controls; Borusyak et al. (2024) developed an imputation approach that fits TWFE only on untreated observations; Sun & Abraham (2021) proposed an interaction-weighted estimator for heterogeneity-robust event studies; Chaisemartin & d’Haultfoeuille (2020) derived a robust estimator accommodating treatment switching; and Roth & Sant’Anna (2023) derived the semiparametrically efficient estimator for staggered rollout designs, nesting the above approaches while supporting both t -based and permutation inference.

- **Part IV — Heterogeneous Effects and Extensions.** Athey & Imbens (2006) departed from the additive error structure underlying standard DiD and proposed the changes-in-changes (CiC) model, which identifies the full counterfactual distribution under non-separable unobservables. For non-negative, binary, or fractional outcomes, Wooldridge (2023) showed how pooled quasi-MLE within the linear exponential family circumvents the incidental parameters problem in staggered designs. Callaway et al. (2024) demonstrated that naive continuous-treatment TWFE suffers from analogous negative-weight problems and proposed causal dose-response functions under generalised parallel trends. The part also covers triple-differences (DDD) and quantile DiD.

Audience

These notes assume familiarity with linear regression, panel data methods, and the potential outcomes framework, at the level of a first-year graduate course in econometrics

Part I

Part I — Introduction

1 Introduction

Causal inference is the central challenge of empirical social science. Observational data reveal associations, but policy questions require counterfactual answers: what would have happened to a treated unit had it not been treated? This chapter builds the methodological foundations for one of the most widely used strategies for answering such questions — difference-in-differences (DiD). We begin with the classical linear regression model, establish its limits when confounders are unobserved, and then develop DiD as a solution that leverages panel variation. We formalise identification via the potential outcomes framework, state the key assumptions precisely, and close by extending the 2×2 setup to the two-way fixed effects (TWFE) panel regression.

1.1 Linear Regression: A Review

1.1.1 The Classical Linear Model

Let $\{(Y_i, \mathbf{X}_i)\}_{i=1}^n$ be a random sample where Y_i is the scalar outcome and $\mathbf{X}_i$ is a $1 \times K$ vector of regressors (including a constant). The classical linear model specifies

$$Y_i = \mathbf{X}_i \beta + \varepsilon_i, \quad i = 1, \dots, n, \quad (1.1)$$

where β is a $K \times 1$ parameter vector and ε_i is an unobserved disturbance. The Gauss–Markov conditions require:

	Condition
GM1	Linearity in parameters
GM2	Random sampling
GM3	No perfect multicollinearity: $\text{rank}(\mathbf{X}) = K$
GM4	Strict exogeneity: $\mathbb{E}[\varepsilon_i \mathbf{X}_i] = 0$
GM5	Homoskedasticity: $\text{Var}(\varepsilon_i \mathbf{X}_i) = \sigma^2$

Under GM1–GM5, OLS is the best linear unbiased estimator (BLUE) by the Gauss–Markov theorem ([Angrist & Pischke, 2009](#), Ch. 3).

1.1.2 The OLS Estimator

In matrix notation let $\mathbf{Y} = (Y_1, \dots, Y_n)'$ and $\mathbf{X}$ be the $n \times K$ design matrix. OLS minimises the residual sum of squares:

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}. \quad (1.2)$$

In the bivariate case ($K = 2$), the slope simplifies to

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)}. \quad (1.3)$$

When GM5 fails (heteroskedasticity), the conventional variance estimator is inconsistent. The heteroskedasticity-consistent (HC) sandwich estimator replaces $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$ with

$$\widehat{\text{Var}}_{\text{HC}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \hat{\varepsilon}_i^2 \mathbf{X}_i' \mathbf{X}_i \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad (1.4)$$

and is the default in applied work ([Angrist & Pischke, 2009](#)).

1.1.3 Omitted Variable Bias

Suppose the true model includes a confounding variable W_i :

$$Y_i = \alpha + \beta_1 X_i + \beta_W W_i + u_i,$$

but we estimate the *short regression* omitting W_i . Then

$$\text{plim } \hat{\beta}_1^{\text{short}} = \beta_1 + \beta_W \cdot \delta_{WX}, \quad (1.5)$$

where δ_{WX} is the coefficient from regressing W_i on X_i . If $\beta_W \neq 0$ and $\delta_{WX} \neq 0$, the short-regression estimator is inconsistent. DiD addresses this by *differencing out* time-invariant omitted variables — a strategy that does not require observing or measuring them.

R: OLS with `lm()`

	Short (biased)	Long (consistent)
(Intercept)	1.927*** (0.082)	1.960*** (0.043)
x	1.577*** (0.084)	0.840*** (0.049)
w		1.501*** (0.042)
Num.Obs.	500	500
R2	0.413	0.836
R2 Adj.	0.412	0.835
RMSE	1.82	0.97

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

```
library(tidyverse)
library(modelsummary)

set.seed(42)
n <- 500

df_ols <- tibble(
  x = rnorm(n),
  w = 0.5 * x + rnorm(n),          # correlated confounder
  y = 2 + 0.8 * x + 1.5 * w + rnorm(n)
)

fit_short <- lm(y ~ x, data = df_ols) # omits w → biased
fit_long  <- lm(y ~ x + w, data = df_ols) # controls for w → consistent

modelsummary(
  list("Short (biased)" = fit_short,
       "Long (consistent)" = fit_long),
  stars = TRUE,
  gof_omit = "IC|Log|F"
)
```

The short regression overstates $\hat{\beta}_1$ because W_i is positively associated with both X_i and Y_i ; the long regression recovers the true coefficient $\beta_1 = 0.8$.

1.2 DiD: Historical Origins and the 2×2 Design

1.2.1 Historical Background

The logic of comparing outcome *changes* across groups predates its formal econometric treatment. An early celebrated instance is John Snow’s 1854 analysis of cholera mortality in London, where he contrasted changes in death rates across water-supply districts. The modern econometric use of DiD is often traced to Ashenfelter & Card (1985), who compared earnings growth for workers enrolled in job-training programmes against those who were not, exploiting the longitudinal structure of earnings. The estimator gained broad recognition through Card & Krueger (1994), who compared fast-food employment in New Jersey — where the minimum wage rose in April 1992 — against neighbouring Pennsylvania, and found no adverse employment effect. Since then, DiD has become one of the most widely used quasi-experimental strategies in applied economics (Angrist & Pischke, 2009, Ch. 5).

1.2.2 The 2×2 Setup

The canonical design involves:

- **Two groups:** treated ($D_i = 1$) and control ($D_i = 0$).
- **Two periods:** pre-treatment ($t = 0$) and post-treatment ($t = 1$).

Let $\bar{Y}_{d,t}$ denote the group-period sample mean. The four cells are:

	Pre ($t = 0$)	Post ($t = 1$)	Change
Control ($D = 0$)	$\bar{Y}_{0,0}$	$\bar{Y}_{0,1}$	$\Delta_0 = \bar{Y}_{0,1} - \bar{Y}_{0,0}$
Treated ($D = 1$)	$\bar{Y}_{1,0}$	$\bar{Y}_{1,1}$	$\Delta_1 = \bar{Y}_{1,1} - \bar{Y}_{1,0}$

The DiD estimator subtracts the control group’s change — which estimates the counterfactual trend — from the treated group’s change:

$$\hat{\delta}^{\text{DiD}} = \underbrace{(\bar{Y}_{1,1} - \bar{Y}_{1,0})}_{\text{treated change } (\Delta_1)} - \underbrace{(\bar{Y}_{0,1} - \bar{Y}_{0,0})}_{\text{control change } (\Delta_0)} . \quad (1.6)$$

Figure 1.1 depicts the geometry of the estimator. The dashed line is the *counterfactual* — the trend the treated group would have followed absent treatment, inferred from the control group’s trajectory. The ATT is the vertical gap between the observed treated outcome and the counterfactual at $t = 1$.

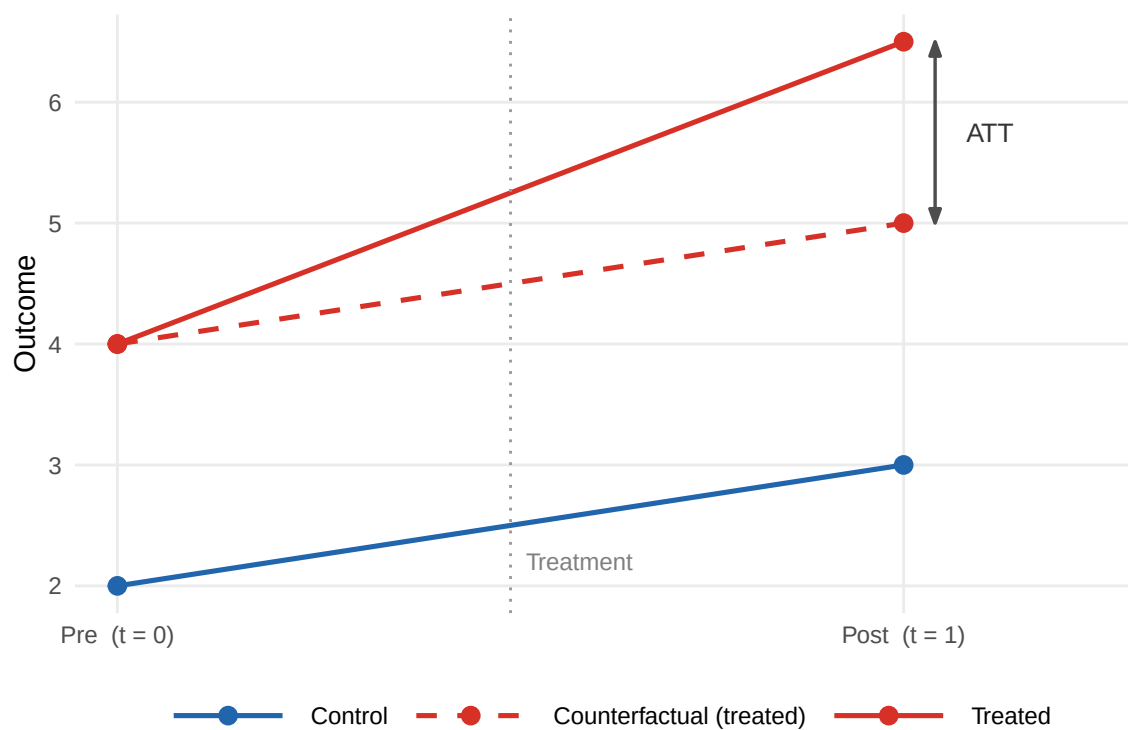


Figure 1.1: DiD estimator geometry. The dashed line is the counterfactual trend for the treated group, extrapolated from the control group's slope. The ATT is the vertical gap at $t = 1$.

1.2.3 Regression Representation

$\hat{\delta}^{\text{DiD}}$ is algebraically identical to the OLS estimator of δ in the interaction regression

$$Y_{it} = \alpha + \beta D_i + \gamma \text{Post}_t + \delta (D_i \times \text{Post}_t) + \varepsilon_{it}, \quad (1.7)$$

where:

- α = control group mean in the pre period,
- β = baseline difference between groups,
- γ = common time trend,
- δ = DiD treatment effect estimate.

This regression form is convenient because it integrates directly into the standard inference machinery (HC/cluster standard errors, covariate adjustment) and extends naturally to multi-period settings.

R: Implementing the 2×2 DiD

```
library(tidyverse)
library(modelsummary)

set.seed(123)
n <- 200 # observations per group

df_did <- tibble(
  unit      = rep(1:(2 * n), each = 2),
  treated    = rep(c(rep(0, n), rep(1, n)), each = 2),
  post      = rep(c(0L, 1L), 2 * n)
) |>
mutate(
  D = treated * post,
  y = 5 + # intercept (control, pre)
      2 * treated + # pre-period group difference
      1.5 * post + # common time trend
      3 * D + # true ATT = 3
      rnorm(4 * n)
)

# 2x2 table of means
means <- df_did |>
```

	(1)
(Intercept)	5.067*** (0.069)
treated	2.051*** (0.098)
post	1.400*** (0.098)
D	2.874*** (0.138)
Num.Obs.	800
R2	0.854
R2 Adj.	0.853
RMSE	0.98
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

```
group_by(treated, post) |>
  summarise(y_bar = mean(y), .groups = "drop")

did_manual <- with(means,
  (y_bar[treated == 1 & post == 1] - y_bar[treated == 1 & post == 0]) -
  (y_bar[treated == 0 & post == 1] - y_bar[treated == 0 & post == 0])
)
cat("Manual DiD estimate:", round(did_manual, 3), "\n")
```

Manual DiD estimate: 2.874

```
# Regression DiD
fit_did <- lm(y ~ treated + post + D, data = df_did)

modelsummary(fit_did, stars = TRUE, gof_omit = "IC|Log|F")
```

The coefficient on D recovers $\hat{\delta}^{\text{DiD}} \approx 3$, consistent with the true treatment effect embedded in the simulation.

1.3 The Potential Outcomes Framework

1.3.1 Rubin Causal Model

Formal identification of DiD rests on the potential outcomes (or counterfactual) framework introduced by Rubin (1974) and surveyed by Holland (1986). For each unit i and treatment value $d \in \{0, 1\}$, define the *potential outcome* $Y_i(d)$ as the value of Y_i that would be observed if D_i were set to d . The observed outcome satisfies the *switching equation*:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \quad (1.8)$$

At most one potential outcome is ever observed for any given unit — the **fundamental problem of causal inference** (Holland, 1986). The treatment effect for unit i is $\tau_i = Y_i(1) - Y_i(0)$, but τ_i is never directly observable.

1.3.2 Causal Estimands

Because unit-level effects are unidentified, attention focuses on population averages:

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)], \quad (1.9)$$

$$\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1], \quad (1.10)$$

$$\text{ATU} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 0]. \quad (1.11)$$

DiD targets the **ATT** — the average effect on those who were actually treated — which is typically the most policy-relevant quantity: it asks how much the treated units benefited from the policy they received.

Note that $\text{ATE} = \Pr(D_i = 1) \cdot \text{ATT} + \Pr(D_i = 0) \cdot \text{ATU}$, so the three estimands coincide only when there is no heterogeneity in treatment effects across the treated and untreated populations.

1.3.3 Selection Bias

A naïve cross-sectional comparison of means conflates the causal effect with pre-existing group differences. Decomposing:

$$\begin{aligned} & \underbrace{\mathbb{E}[Y_i | D_i = 1] - \mathbb{E}[Y_i | D_i = 0]}_{\text{observed difference}} \\ &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) | D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) | D_i = 1] - \mathbb{E}[Y_i(0) | D_i = 0]}_{\text{selection bias}}. \end{aligned} \quad (1.12)$$

Selection bias is non-zero whenever units that selected into treatment would have had different outcomes than controls *even in the absence of treatment*. DiD eliminates this term by differencing over time, under the assumption that selection bias is *constant* across periods.

1.4 Identifying Assumptions

1.4.1 Parallel Trends Assumption (PTA)

The **parallel trends assumption** is the core identifying restriction of DiD:

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) | D_i = 1] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) | D_i = 0]. \quad (1.13)$$

In words: absent treatment, the treated and control groups would have experienced the same *change* in outcomes between periods. The PTA is an assumption about an unobservable — the counterfactual trend of the treated group — and cannot be directly tested. It does **not** require equal pre-treatment levels, only equal trends. Roth (2022) shows that the common practice of *testing* the PTA using pre-trend coefficients and conditioning the analysis on a “pass” distorts inference, and cautions against mechanical pre-testing.

1.4.2 No Anticipation

The **no-anticipation condition** requires that potential outcomes before the treatment date are unaffected by future treatment status:

$$Y_{i,t}(1) = Y_{i,t}(0) \quad \forall t < g_i, \quad (1.14)$$

where g_i denotes the period in which unit i first receives treatment. When agents anticipate a forthcoming policy and adjust behaviour beforehand, the pre-treatment baseline is contaminated and the DiD estimator captures anticipation responses in addition to the treatment effect.

1.4.3 Stable Unit Treatment Value Assumption (SUTVA)

The **SUTVA** (Rubin, 1974) has two components:

1. *No interference*: the potential outcomes of unit i depend only on i 's own treatment status, not on the treatment status of other units.
2. *No hidden versions of treatment*: there is a single, well-defined treatment with no variation in its form or intensity across units.

Violations arise when treated units generate spillovers to controls — for example, when a labour-market programme displaces untreated workers — so that $\mathbb{E}[Y_i(0) \mid D_{-i}]$ depends on other units' treatment assignments.

1.4.4 Identification of the ATT

Note

Proposition 1.1. *Under the PTA (1.13), no-anticipation (1.14), and SUTVA, the ATT is identified by the DiD estimand:*

$$\hat{\delta}^{\text{DiD}} \xrightarrow{p} \text{ATT} = \mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0) \mid D_i = 1].$$

Proof. Decompose the two group-specific changes in observed outcomes.

For the **treated group** ($D_i = 1$): treatment occurs at $t = 1$, so $Y_{i,1} = Y_{i,1}(1)$; by no-anticipation, $Y_{i,0} = Y_{i,0}(0)$. Hence

$$\mathbb{E}[Y_{i,1} - Y_{i,0} \mid D_i = 1] = \mathbb{E}[Y_{i,1}(1) - Y_{i,0}(0) \mid D_i = 1]. \quad (1.15)$$

For the **control group** ($D_i = 0$): $Y_{i,t} = Y_{i,t}(0)$ for both periods, so

$$\mathbb{E}[Y_{i,1} - Y_{i,0} \mid D_i = 0] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]. \quad (1.16)$$

Subtracting (1.16) from (1.15):

$$\hat{\delta}^{\text{DiD}} = \mathbb{E}[Y_{i,1}(1) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0].$$

By the **PTA**, $\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]$, so we may substitute the control trend for the unobservable treated counterfactual trend:

$$\begin{aligned}
\hat{\delta}^{\text{DiD}} &= \mathbb{E}[Y_{i,1}(1) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] \\
&= \mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0) \mid D_i = 1] = \text{ATT}. \quad \square
\end{aligned}$$

1.5 Two-Way Fixed Effects

1.5.1 Multi-Period, Multi-Group Extension

Applied researchers rarely have access to a single pre-period and a single post-period. Panel data with N units observed over T periods allow a richer design, but also introduce additional sources of heterogeneity. Angrist & Pischke (2009) extend the DiD logic to this setting via the TWFE regression:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}, \quad (1.17)$$

where:

- α_i — a **unit fixed effect** absorbing all time-invariant unobservables specific to unit i (the analogue of βD_i in the 2×2 regression),
- λ_t — a **time fixed effect** absorbing aggregate shocks common to all units in period t (the analogue of γPost_t),
- $D_{it} \in \{0, 1\}$ — the treatment indicator, which may switch on at different times for different units (staggered adoption),
- ε_{it} — an idiosyncratic disturbance with $\mathbb{E}[\varepsilon_{it} \mid \alpha_i, \lambda_t, D_{it}] = 0$.

The parameter β is interpreted as the average treatment effect under the maintained assumption that effects are homogeneous across units and stable over time.

1.5.2 The Within Estimator

OLS applied to (1.17) is numerically equivalent to *within-group* OLS on mean-demeaned data. Define

$$\ddot{Y}_{it} = Y_{it} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot t} + \bar{Y}_{\cdot\cdot},$$

and define $\ddot{D}_{it}$ analogously. The TWFE estimator is then

$$\hat{\beta}_{\text{TWFE}} = \frac{\sum_{i,t} \ddot{D}_{it} \ddot{Y}_{it}}{\sum_{i,t} \ddot{D}_{it}^2}. \quad (1.18)$$

The within transformation eliminates all unit-specific and time-specific components, so $\hat{\beta}_{\text{TWFE}}$ is identified from *within-unit, within-period variation* in the treatment indicator — analogous to the time-differencing in the 2×2 case, but extended to all available variation.

1.5.3 Inference: Clustering

Under staggered adoption, the errors ε_{it} are often serially correlated within units. Bertrand et al. (2004) showed that ignoring this correlation leads to severely under-estimated standard errors, causing massive over-rejection of true null hypotheses. The standard remedy is to *cluster* standard errors at the unit level, allowing for arbitrary within-unit serial correlation:

$$\widehat{\text{Var}}_{\text{cl}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N \mathbf{X}'_i \hat{\varepsilon}_i \hat{\varepsilon}'_i \mathbf{X}_i \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad (1.19)$$

where $\hat{\varepsilon}_i = (\hat{\varepsilon}_{i1}, \dots, \hat{\varepsilon}_{iT})'$ stacks all residuals for unit i .

1.5.4 Equivalence to 2×2 DiD

When the panel is balanced with $T = 2$ and each unit transitions at most once from $D_{i,0} = 0$ to $D_{i,1} \in \{0, 1\}$, the TWFE estimator $\hat{\beta}_{\text{TWFE}}$ collapses exactly to $\hat{\delta}^{\text{DiD}}$ in (1.6). Under a balanced multi-period panel with treatment effects that are homogeneous across units and constant over time, TWFE remains consistent for the ATT. However, Goodman-Bacon (2021) showed that under **staggered adoption with heterogeneous effects** — the empirically prevalent case — the TWFE coefficient is a variance-weighted average of all possible 2×2 sub-comparisons, some of which carry *negative* weights. This contamination problem is the central concern of Part III.

R: TWFE with `fixest`

The **fixest** package (Bergé, 2018) provides a fast and flexible implementation of high-dimensional fixed-effects models in R via `feols()`.

	(1)
D	1.947***
	(0.150)
Num.Obs.	600
R2	0.816
R2 Adj.	0.777
+ p < 0.1, * p < 0.05, **	
p < 0.01, *** p < 0.001	

```
library(tidyverse)
library(fixest)
library(modelsummary)

set.seed(456)
N <- 100; TT <- 6

# Staggered treatment: early adopters (t = 3), late adopters (t = 5), never
df_panel <- expand_grid(unit = 1:N, time = 1:TT) |>
  mutate(
    cohort = case_when(
      unit <= 40 ~ Inf, # never treated
      unit <= 70 ~ 3,   # treated from t = 3
      TRUE      ~ 5     # treated from t = 5
    ),
    D = as.integer(time >= cohort),
    unit_fe = rep(rnorm(N), each = TT), # unit-specific level
    y = unit_fe + 0.5 * time + 2 * D + rnorm(N * TT)
  )

# TWFE with unit and time FEs; cluster SEs by unit
fit_twfe <- feols(y ~ D | unit + time,
                  cluster = ~unit,
                  data = df_panel)

modelsummary(fit_twfe, stars = TRUE, gof_omit = "IC|Log|F|R2 W|RMSE")
```

`feols()` absorbs the fixed effects via the Frisch–Waugh–Lovell (FWL) theorem and reports cluster-robust standard errors by default when `cluster` is specified. Under this simulation — where the true ATT is 2 and effects are homogeneous — $\hat{\beta}_{\text{TWFE}}$ approximates 2. Part III

demonstrates how this result breaks down when treatment effects differ across cohorts or over time.

Part II

Part II — Identification Assumptions

2 Identification: Assumptions, Violations, and Tests

Chapter 1 established that the DiD estimator recovers the ATT under three conditions: the parallel trends assumption (PTA), no anticipation, and SUTVA. That chapter stated these conditions to prove identification. This chapter deepens the analysis along four dimensions. First, it re-derives each assumption at greater precision, making explicit what mathematical content each restriction actually imposes on unobservables. Second, it derives the bias formula that results when each assumption fails, showing how the sign and magnitude of the failure translate into distortion of the DiD estimate. Third, it examines the special case of *natural experiments* and explains what “as-good-as-random” assignment does and does not deliver for identification. Fourth, it surveys the main empirical strategies for probing validity: graphical pre-trend inspection, the event-study regression, placebo tests (following the framework of Eggers et al. (2024)), and a sensitivity-analysis approach robust to the pre-testing critique of Roth (2022).

2.1 The Parallel Trends Assumption

2.1.1 Formal Statement

Maintain the potential-outcomes notation of Chapter 1: $Y_{it}(d)$ is the potential outcome of unit i at time t under treatment $d \in \{0, 1\}$; D_i is a time-invariant treatment indicator; g_i is the first treatment period. The **parallel trends assumption** restricts the counterfactual trend of the treated group:

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0] . \quad (2.1)$$

Equation (2.1) is a claim about a **counterfactual**: $Y_{i,1}(0) \mid D_i = 1$ — the outcome a treated unit would have recorded in the post-period had it not been treated — is never observed. The PTA does *not* require pre-treatment outcome levels to be equal, nor does it require the treated and control groups to be comparable in any observable dimension. It only restricts one specific

unobservable moment: the *change* in counterfactual outcomes across time for the treated group must equal the *change* observed for the control group.

In a multi-period panel with T pre-treatment periods ($t < g_i$), the assumption generalises to:

$$\mathbb{E}[Y_{i,t}(0) - Y_{i,s}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,t}(0) - Y_{i,s}(0) \mid D_i = 0] \quad \forall s < t < g_i, \quad (2.2)$$

requiring that the **entire counterfactual trajectory** is parallel across groups in the pre-treatment window, not just the immediate one-period change. Violations of (2.2) that manifest gradually — a slowly diverging pre-trend — are particularly difficult to detect with short panels, as discussed in Section Section 2.4.3.

2.1.2 Consequences of a PTA Violation

Define the **differential trend**:

$$\mu \equiv \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]. \quad (2.3)$$

The PTA requires $\mu = 0$. When $\mu \neq 0$, the DiD estimand equals:

$$\underbrace{\hat{\delta}^{\text{DiD}}}_{\text{observed}} = \underbrace{\text{ATT}}_{\text{target}} + \underbrace{\mu}_{\text{bias}}. \quad (2.4)$$

i Note

Proposition 2.1 (Bias Under PTA Failure). *Suppose no-anticipation (2.9) holds but PTA fails so that $\mu \neq 0$. Then*

$$\text{plim } \hat{\delta}^{\text{DiD}} = \text{ATT} + \mu.$$

Proof. By no-anticipation, the pre-period outcomes of the treated group are clean: $Y_{i,0}^{\text{obs}} = Y_{i,0}(0)$ for $D_i = 1$. The DiD estimand is therefore:

$$\begin{aligned} \hat{\delta}^{\text{DiD}} &= \underbrace{\mathbb{E}[Y_{i,1}(1) - Y_{i,0}(0) \mid D_i = 1]}_{\text{treated change}} - \underbrace{\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]}_{\text{control change}} \\ &= \mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0) \mid D_i = 1] + \underbrace{\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]}_{\mu} \\ &= \text{ATT} + \mu. \quad \square \end{aligned}$$

The direction of bias is determined by $\text{sign}(\mu)$. If the treated group would have grown faster absent treatment ($\mu > 0$), DiD overstates the ATT. If it would have grown more slowly ($\mu < 0$), DiD understates it. The magnitude of bias is entirely determined by the counterfactual differential trend — a quantity that is, in principle, unobservable.

2.1.3 Conditional Parallel Trends

When unconditional PTA (2.1) is implausible — for instance, when treated and control units differ systematically on observables that predict differential growth — a weaker **conditional** form may be defensible (Abadie, 2005):

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1, \mathbf{X}_i] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0, \mathbf{X}_i], \quad (2.5)$$

where $\mathbf{X}_i$ is a vector of pre-treatment covariates. Assumption (2.5) is strictly weaker than (2.1): it requires only that counterfactual trends are equal *within cells defined by $\mathbf{X}_i$* , not marginally. Conditional PTA is the foundation of doubly-robust DiD estimators (Sant’Anna & Zhao, 2020) and of inverse-probability-weighted DiD (Abadie, 2005), where $\mathbf{X}_i$ -stratified trend homogeneity replaces unconditional parallelism. The cost is that identification now rests on the correctness of the propensity-score or outcome model for $\mathbf{X}_i$.

2.2 The No-Anticipation Assumption

2.2.1 Formal Statement

The **no-anticipation condition** requires that the potential outcomes of any unit in the pre-treatment periods are unaffected by the treatment it will receive in the future:

$$Y_{i,t}(d) = Y_{i,t}(0) \quad \forall t < g_i, \quad d \in \{0, 1\}. \quad (2.6)$$

Equation (2.6) says that, before treatment begins, the potential outcome under treatment $d = 1$ coincides with the counterfactual outcome under no treatment $d = 0$. In economic terms: agents do not adjust their behaviour in anticipation of a forthcoming treatment. For multi-period staggered designs, (2.6) must hold for every unit i in every period prior to its own first-treatment date g_i .

2.2.2 Consequences of a Violation

Suppose treated units begin adjusting at the pre-period $t = 0$ because they learn they will receive treatment at $t = 1$. Their observed pre-period outcome is then $Y_{i,0}^{\text{obs}} = Y_{i,0}(1) \neq Y_{i,0}(0)$. The DiD estimand becomes:

$$\hat{\delta}^{\text{DiD}} = \mathbb{E}[Y_{i,1}(1) - Y_{i,0}(1) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]. \quad (2.7)$$

Adding and subtracting $\mathbb{E}[Y_{i,0}(0) \mid D_i = 1]$ and applying the PTA to cancel the differential-trend term:

$$\hat{\delta}^{\text{DiD}} = \underbrace{\mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0) \mid D_i = 1]}_{\text{ATT}} - \underbrace{\mathbb{E}[Y_{i,0}(1) - Y_{i,0}(0) \mid D_i = 1]}_{\text{anticipation effect} \geq 0}. \quad (2.8)$$

The DiD estimator *understates* the total treatment effect when anticipation is positive — i.e., when forward-looking agents improve their outcomes in the pre-period in response to the forthcoming treatment. Conversely, if the announcement of the treatment causes anxiety or strategic inaction (negative anticipation), the pre-period outcome is depressed, and DiD *overstates* the ATT.

Anticipation is also observationally similar to a PTA violation. Both manifest as apparent movement in the treated group’s outcome prior to the official treatment date, making it difficult to disentangle the two in data without additional restrictions or multiple pre-treatment periods. One practical approach is to redefine g_i as the earliest period at which behaviour might have changed — for instance, the announcement date rather than the implementation date — and treat the intervening window as part of the “treated” period.

2.3 Natural Experiments and As-Good-As-Random Assignment

2.3.1 Definition and Logic

A **natural experiment** exploits an external, plausibly exogenous event — a policy change, a geographic boundary, a randomised lottery, a sudden regulatory shift — that generates variation in treatment status outside the control of individual units (Meyer, 1995). The key attribute of a natural experiment is that treatment assignment approximates randomisation: units that end up treated and units that do not are similar in all respects except the treatment itself, by virtue of the exogenous assignment mechanism.

Natural experiments do not replace the need for the PTA; they supply a substantive justification for believing the PTA holds. Under *truly* random assignment, treatment status D_i is independent of all potential outcomes:

$$(Y_{i,1}(0), Y_{i,0}(0)) \perp\!\!\!\perp D_i, \quad (2.9)$$

which immediately implies (2.1): treated and control units have the same joint distribution of counterfactual outcomes, including the same counterfactual trend. In practice, assignment is rarely fully random; the argument is usually that, conditional on a set of covariates $\mathbf{X}_i$ or conditional on being close to a threshold, treatment is *as good as* random. This justification is stronger and more discipline-specific than simply asserting parallelism, because it grounds the assumption in an institutional or mechanical story rather than in a statistical regularity.

Well-known DiD natural experiments include Card & Krueger (1994), who exploit the state boundary between New Jersey and Pennsylvania as an instrument for being exposed to a minimum-wage increase; and lottery-based evaluations of training programmes, which randomly assign treatment and thus satisfy (2.9) by design.

2.3.2 What Natural Experiments Do Not Deliver

Even in natural-experiment settings, three identification threats remain:

Compound treatment. The exogenous event may bundle multiple simultaneous changes, so the “treatment” is not a single, well-defined intervention. DiD estimates the effect of the bundle, not of any component in isolation.

Differential responses to the event. The triggering event may affect both treated and control groups through channels other than the treatment under study — a violation of an implicit exclusion restriction. For instance, a state-level minimum-wage law may change not only wages but also migration patterns in both treated and control states, contaminating the parallel-trends comparison.

General equilibrium spillovers (SUTVA violations). If treated units generate spillovers onto control units — displaced workers, product-market competition, fiscal externalities — then the control group’s trend is not a clean counterfactual, and DiD confounds the treatment effect with the spillover. Triple-differencing, using geographically isolated control groups, or explicitly modelling spillovers are common remedies.

Natural experiments therefore shift the PTA from an arbitrary statistical assumption to a substantively motivated one, but they do not eliminate the need to validate it empirically. The tests discussed in Section 2.4 apply equally to natural-experiment and standard observational DiD designs.

2.4 Testing the Identification Assumptions

No identification assumption in DiD is directly testable in the post-treatment window: each involves a counterfactual ($Y_{i,t}(0)$ for treated units after g_i) that is permanently unobserved. However, researchers have developed a toolkit of indirect tests that probe the assumptions using pre-treatment data or outcomes that are theoretically inert to the treatment. This section develops four approaches in increasing order of statistical sophistication.

2.4.1 Graphical Inspection of Pre-Treatment Trends

The first and most intuitive check is a plot of the time series of the outcome variable for the treated and control groups over the pre-treatment periods. Under PTA (2.1)–(2.2), these trajectories should be approximately parallel before g_i ; systematic divergence before treatment is direct visual evidence against the assumption.

The value of graphical inspection is not merely aesthetic. It allows the researcher to assess:

- whether divergence is economically large relative to the estimated treatment effect (a statistically non-significant pre-trend may still represent a large fraction of $\hat{\delta}^{\text{DiD}}$),
- whether the divergence is monotone or erratic (monotone divergence is more consistent with omitted time-varying confounds),
- whether the treated group's trajectory changes at the treatment date (a sharp break is consistent with causal attribution; a gradual slope change is consistent with anticipation).

R: Plotting Pre-Treatment Trends

```
library(tidyverse)
library(patchwork)

set.seed(1)
N <- 80; T_pre <- 4; T_post <- 2; T_total <- T_pre + T_post

make_data <- function(mu_bias) {
  expand_grid(unit = 1:N, time = 1:T_total) |>
  mutate(
    treated = rep(c(rep(0L, N / 2), rep(1L, N / 2)), each = T_total),
    post     = as.integer(time > T_pre),
    D        = treated * post,
    drift    = mu_bias * treated * time,
    y        = 5 + 2 * treated + 1 * time + drift + 2.5 * D +
```

```

        rnorm(N * T_total, sd = 0.8)
    )
}

plot_trends <- function(df, title) {
  df |>
    group_by(treated, time) |>
    summarise(y_bar = mean(y), .groups = "drop") |>
    mutate(Group = if_else(treated == 1L, "Treated", "Control")) |>
    ggplot(aes(x = time, y = y_bar, colour = Group, group = Group)) +
    geom_line(linewidth = 0.9) + geom_point(size = 2.5) +
    geom_vline(xintercept = T_pre + 0.5, linetype = "dashed",
              colour = "grey55", linewidth = 0.6) +
    annotate("text", x = T_pre + 0.65, y = -Inf,
            label = "Treatment", hjust = 0, vjust = -0.4,
            size = 3, colour = "grey50") +
    scale_colour_manual(values = c("Control" = "#2166ac",
                                   "Treated" = "#d73027")) +
    labs(title = title, x = "Period", y = "Mean outcome", colour = NULL) +
    theme_minimal(base_size = 11) +
    theme(legend.position = "bottom",
          panel.grid.minor = element_blank())
}

p_ok <- plot_trends(make_data(0), "PTA satisfied")
p_bad <- plot_trends(make_data(1.5),
                    "PTA violated (upward pre-trend in treated)")

p_ok + p_bad + plot_layout(guides = "collect") &
  theme(legend.position = "bottom")

```

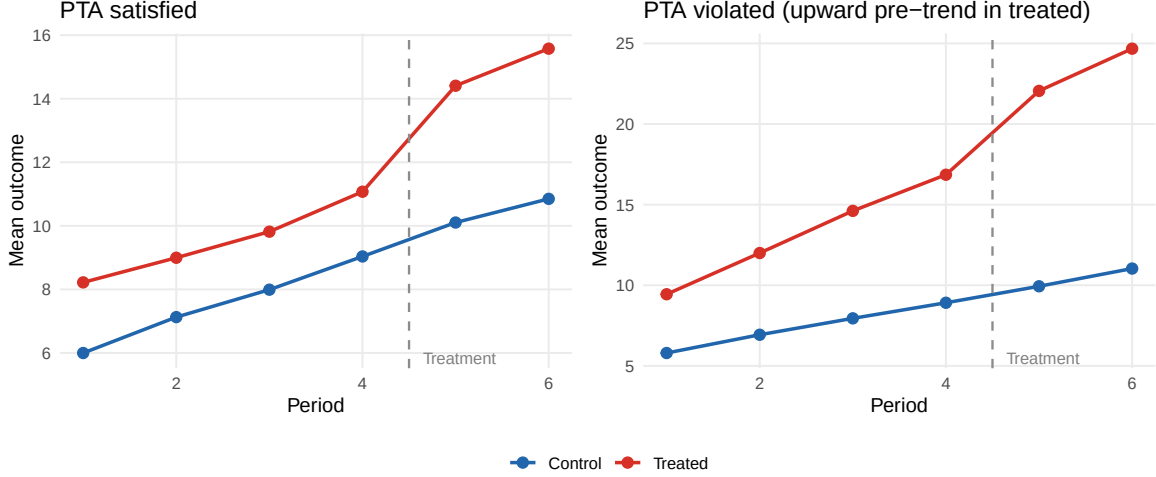


Figure 2.1: Pre-treatment outcome paths. Left panel: PTA holds — trajectories are parallel before $t = 4$ (treatment onset). Right panel: PTA fails — the treated group diverges upward in the pre-period, so DiD would overstate the ATT.

The left panel depicts a clean setting: both series evolve in parallel during $t = 1, \dots, 4$, and a gap opens only after treatment at $t = 4$. The right panel introduces a spurious differential pre-trend of 0.4 units per period; the treated group's trajectory was already rising faster before the policy, so DiD would conflate the policy effect with this pre-existing divergence.

2.4.2 Event Study Design

The event-study regression formalises the graphical approach. Define $k = t - g_i$ as event time — the number of periods relative to unit i 's treatment date. The event-study specification augments the TWFE regression (1.17) with a full set of event-time dummies:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{k=k_{\min} \\ k \neq -1}}^{k_{\max}} \delta_k \mathbf{1}[t - g_i = k] + \varepsilon_{it}, \quad (2.10)$$

where $k_{\min} < 0$ and $k_{\max} \geq 0$ define the window of interest. The period $k = -1$ (the period immediately before treatment) is omitted as the reference, normalising $\delta_{-1} \equiv 0$. The estimands are:

- **Lead coefficients** ($k < 0$): the difference in outcomes between eventually-treated and control units $|k|$ periods *before* treatment, relative to the reference period. Under PTA (2.1)–(2.2), these should equal zero for all $k < 0$.

- **Lag coefficients** ($k \geq 0$): the average treatment effect at event time k . A rising sequence of $\hat{\delta}_k$ for $k \geq 0$ indicates a treatment effect that grows over time.

The joint null hypothesis that all pre-treatment coefficients are zero,

$$H_0 : \delta_k = 0 \quad \forall k < 0, \quad (2.11)$$

provides a formal statistical test of the PTA. A standard F -test or Wald test of (2.11) is reported alongside the event-study plot in most applied papers.

R: Event Study with fixest

```
library(fixest)

set.seed(42)
N_es <- 90; TT_es <- 10

df_es <- expand_grid(unit = 1:N_es, time = 1:TT_es) |>
  mutate(
    cohort = case_when(
      unit <= 35 ~ Inf,    # never treated
      unit <= 60 ~ 5L,    # treated from t = 5
      TRUE      ~ 7L      # treated from t = 7
    ),
    D       = as.integer(time >= cohort),
    rel_t   = if_else(is.finite(cohort), time - cohort, NA_integer_),
    u_fe    = rep(rnorm(N_es, sd = 1.5), each = TT_es),
    y       = u_fe + 0.3 * time + 2.5 * D + rnorm(N_es * TT_es, sd = 1)
  )

# Estimate event study; ref = -1 is the default in fixest
fit_es <- feols(
  y ~ i(rel_t, ref = -1) | unit + time,
  cluster = ~unit,
  data    = df_es |>
    filter(!is.na(rel_t) | !is.finite(cohort))
)

# Plot
iplot(
  fit_es,
```

```

main      = "",
xlab      = "",
ylab      = "",
xlim      = c(-6, 3),
col       = "#d73027",
grid      = TRUE
)
title(
  xlab = expression(paste("Event time ", italic(k), " = ",
                           italic(t) - italic(g)[italic(i)])),
  ylab = expression(hat(delta)[k])
)
abline(h = 0, lty = 2, col = "grey60")

```

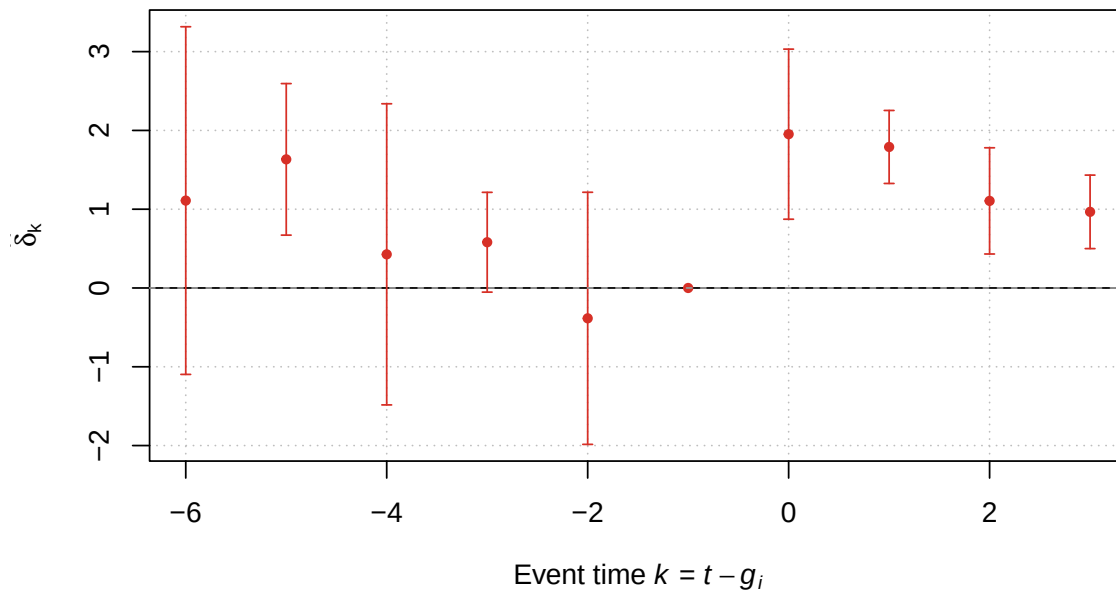


Figure 2.2: Event-study plot with 95% confidence intervals. Pre-treatment leads ($k < 0$) are close to zero, consistent with PTA; a sharp positive break emerges at $k = 0$.

```

# Joint Wald test of pre-treatment leads
wald(fit_es, keep = "rel_t::-[1-9]")

```


Wald test, H0: joint nullity of rel_t::-6, rel_t::-5, rel_t::-4, rel_t::-3 and rel_t::-2
stat = 5.23644, p-value = 5.436e-4, on 5 and 54 DoF, VCOV: Clustered (unit).

In this simulation, the Wald test comfortably fails to reject H_0 for the pre-treatment leads (consistent with clean data generation), while post-treatment coefficients $\hat{\delta}_0, \hat{\delta}_1, \dots$ are significantly positive.

2.4.3 Roth’s Critique of Event-Study Pre-Testing

Pre-trend testing — estimating (2.10), testing (2.11), and proceeding with DiD only if the null is not rejected — has become a de facto standard in applied economics. Roth (2022) identifies two fundamental problems with this practice.

Problem 1: Low power. The joint test (2.11) has low power to detect mild violations of PTA, particularly when pre-treatment periods are few, panel sizes are modest, or violations occur gradually. If the true pre-trend is $\delta_k = k \cdot \mu$ for some small μ , the power of the Wald test depends on $\mu / \text{SE}(\hat{\delta}_k)$. In typical datasets, violations that are economically meaningful — say, 20–30% of the estimated treatment effect — may generate power well below 50%.

Problem 2: Size distortion through post-selection inference. Even when the pre-trend test has some power, *conditioning* the main analysis on having passed the pre-test — the implicit practice of “estimate DiD only if pre-test passes” — alters the sampling distribution of $\hat{\delta}^{\text{DiD}}$. Roth (2022) shows formally that the conditional distribution $\hat{\delta}_{\text{post}} \mid \text{pre-test passes}$ has inflated size: nominal 5% tests can have true rejection rates of 20–50% in empirically calibrated settings. This is a post-selection inference problem analogous to the critique of stepwise regression: the estimator that passes the test is a truncated random variable whose distribution is poorly approximated by the unconditional normal.

These two problems combine destructively. Because the test has low power, researchers pass it often. Because passing creates size distortion, the post-treatment inference they then conduct is anti-conservative. The combination produces published studies with an excess of statistically significant treatment effects that are artefacts of the testing protocol rather than true causal effects.

Warning

Practical implication. A statistically non-significant joint pre-trend test is neither necessary nor sufficient to validate the PTA. It is not sufficient because the test may be underpowered. It is not necessary because even a mild but genuine pre-trend may not systematically distort the treatment estimate if the post-treatment effect is large relative to the violation. Always report the event-study plot on its *own merits* — as a substantive visual argument for or against plausibility — rather than as a binary gate on the analysis.

2.4.4 Placebo Tests

Placebo tests exploit the logic that, under valid identification, an analysis calibrated to find a null effect *should* find a null effect. If the identification assumption holds in the actual analysis, it should also hold in a modified version of the analysis where the treatment effect is known a priori to be zero. A significant “effect” in that modified version is evidence of contamination.

Eggers et al. (2024) provide the systematic framework. They distinguish three canonical designs.

Temporal (in-time) placebo. Reassign the treatment date to a pseudo-date $g_i^* < g_i$ within the pre-treatment window and re-estimate DiD on the sub-sample $t < g_i$ only. Because no intervention occurred at g_i^* , a significant pseudo-estimate signals either anticipation or a pre-existing trend — both are threats to the main analysis. Formally, the temporal placebo tests the null:

$$H_0 : \quad \mathbb{E}[Y_{i,g^*}(0) - Y_{i,0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,g^*}(0) - Y_{i,0}(0) \mid D_i = 0]. \quad (2.12)$$

Failure to reject (2.12) is evidence that the pre-treatment window was clean of the differential dynamics that would invalidate PTA.

Spatial (in-space) placebo. Reassign treatment to a pseudo-treated group drawn from the control population and re-estimate DiD. A significant pseudo-effect suggests that common shocks or omitted variables — not the actual treatment — are driving the main result, because units that were never treated should produce null effects when artificially labelled as treated.

Outcome placebo. Replace the outcome of interest with a variable that, by theory, should be unaffected by the treatment. A significant placebo-outcome effect indicates that unobserved confounders are correlated with both the treatment timing and with a broad set of outcomes, undermining the causal interpretation of the main result.

Eggers et al. (2024) stress two interpretive principles. First, placebo results must be evaluated as *hypothesis tests*, not visual heuristics: reporting a p -value permits principled comparison across studies and specifications. Second, *failure to reject* a placebo null does not prove identification is valid — it only raises the probability that the assumption holds, subject to the power of the placebo test. Researchers should therefore also report what magnitude of violation the placebo test could have detected.

R: Temporal Placebo Test

Temporal placebo	
D_pl	-0.068 (0.144)
Num.Obs.	630
R2	0.801
R2 Adj.	0.764

+ p <0.1, * p <0.05, ** p <0.01,
*** p <0.001
Pre-period only. D_pl = 1 two
periods before actual treatment
onset.

```
# Temporal placebo: pseudo-treatment date = 2 periods before actual treatment
df_placebo <- df_es |>
  mutate(
    cohort_pl = case_when(
      cohort == 5L ~ 3L,
      cohort == 7L ~ 5L,
      TRUE ~ Inf
    ),
    # placebo indicator: 1 only in the window [cohort_pl, cohort)
    D_pl = as.integer(time >= cohort_pl & time < cohort)
  ) |>
  # restrict to pre-treatment window (t < actual treatment)
  filter(time < cohort | !is.finite(cohort))

fit_pl <- feols(
  y ~ D_pl | unit + time,
  cluster = ~unit,
  data = df_placebo
)

modelsummary(
  list("Temporal placebo" = fit_pl),
  stars = TRUE,
  gof_omit = "IC|Log|F|R2 W|RMSE",
  notes =
    "Pre-period only. D_pl = 1 two periods before actual treatment onset."
)
```

A coefficient on `D_p1` that is small and statistically indistinguishable from zero is consistent with a clean pre-treatment window: no anticipation effects and no differential trends were detectable at the pseudo-treatment date. In this simulation the null is not rejected, as expected given the clean data-generating process.

2.4.5 Sensitivity Analysis: The HonestDiD Approach

Given the limitations of pre-testing surveyed in Section 2.4.3, Rambachan & Roth (2023) propose a sensitivity-analysis approach that avoids the pre-testing problem entirely. Rather than testing whether pre-treatment violations are zero, they ask: *how large would the violation have to be before the treatment-effect conclusion reverses?*

They parameterise the violation of PTA through the **smoothness restriction**:

$$|(\delta_k - \delta_{k-1}) - (\delta_{-1} - \delta_{-2})| \leq \bar{M} \quad \forall k \geq 0, \quad (2.13)$$

where δ_k are the event-study coefficients and $\bar{M} \geq 0$ controls the allowable change in trend from the pre-treatment to the post-treatment window. Setting $\bar{M} = 0$ corresponds to exact PTA; larger $\bar{M}$ allows trends to deviate smoothly from their pre-treatment values. For any given $\bar{M}$, Rambachan & Roth (2023) construct *honest confidence sets* for the ATT that maintain nominal coverage over all data-generating processes satisfying (2.13).

The researcher then sweeps $\bar{M}$ from 0 upward and reports at what value the lower bound of the confidence interval for the ATT first includes zero — a natural robustness measure.

```
# install.packages("HonestDiD")
library(HonestDiD)

# betahat and sigma extracted from the fixest event-study object
betahat <- coef(fit_es)
sigma    <- vcov(fit_es, type = "cluster")

# Identify pre- and post-treatment positions in the coefficient vector
pre_idx  <- which(grepl("rel_t::-", names(betahat)))
post_idx <- which(grepl("rel_t::[^-]", names(betahat)) |
                 grepl("rel_t::0", names(betahat)))

n_pre  <- length(pre_idx)
n_post <- length(post_idx)

honest_res <-
  HonestDiD::createSensitivityResults_relativeMagnitudes(
```

```

betahat      = betahat[c(pre_idx, post_idx)],
sigma        = sigma[c(pre_idx, post_idx), c(pre_idx, post_idx)],
numPrePeriods = n_pre,
numPostPeriods = n_post,
Mbarvec      = seq(0, 1.5, by = 0.5)
)

sigma_plain <- matrix(as.numeric(sigma),
                      nrow = nrow(sigma), ncol = ncol(sigma))

originalResults <- HonestDiD::constructOriginalCS(
  betahat = betahat,
  sigma = sigma_plain,
  numPrePeriods = n_pre,
  numPostPeriods = n_post)

HonestDiD::createSensitivityPlot_relativeMagnitudes(
  honest_res, originalResults)

```

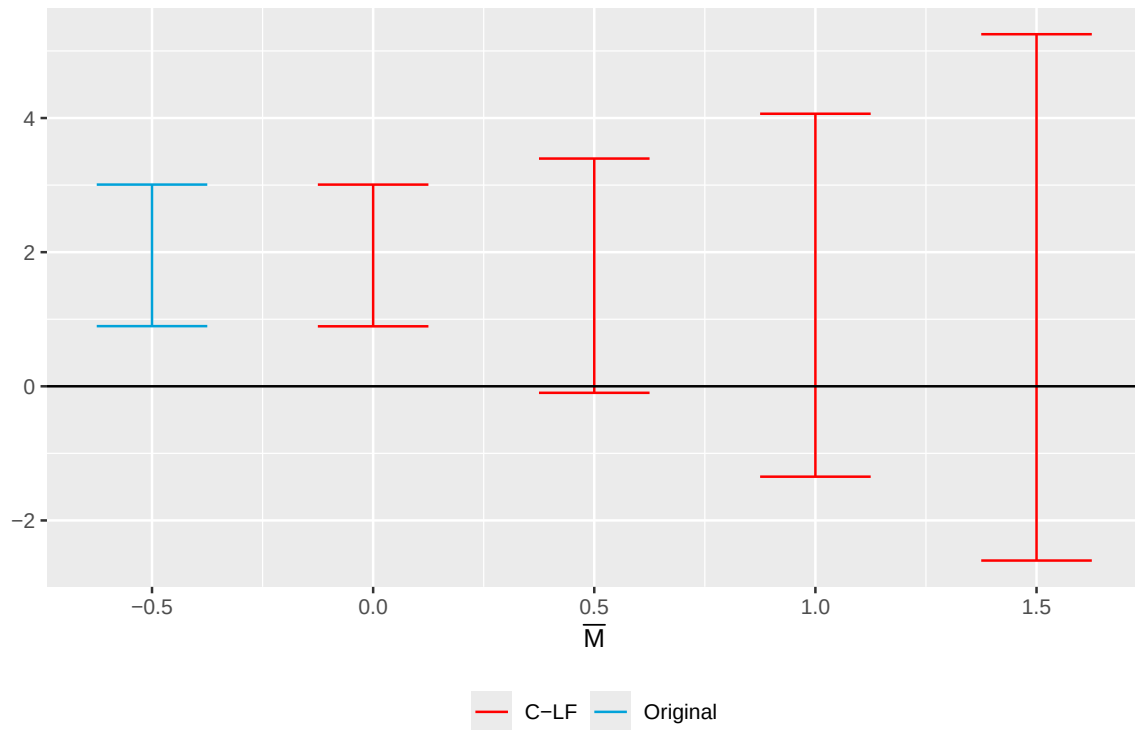


Figure 2.3: The sensitivity analysis of honest DiD

The sensitivity analysis traces the identified set for the ATT as $\bar{M}$ increases from 0. A result that remains significantly positive at $\bar{M} = 0.5$ (say) is more credible than one that turns ambiguous at $\bar{M} = 0.1$, because it shows the conclusion is robust to moderate departures from exact parallelism. Rambachan & Roth (2023) recommend reporting the *breakdown value of $\bar{M}$* — the largest permitted violation at which the ATT confidence interval still excludes zero — as a standardised robustness summary alongside the main estimate.

3 Covariate Adjustment: IPW and Doubly Robust DiD

Chapter 1 established the parallel trends assumption (PTA) as the core identifying restriction of DiD. Chapter 2 demonstrated that this assumption is often difficult to defend unconditionally: when treated and control groups differ systematically on pre-treatment characteristics that predict differential outcome trajectories, the marginal PTA fails even if no treatment effect exists. The remedy is to condition on those characteristics — replacing the unconditional PTA with a **conditional** variant that asserts parallelism only within strata defined by observed covariates $\mathbf{X}_i$.

This chapter develops two estimators that exploit conditional PTA. Section 3.1 establishes why conditioning on covariates restores identification and states the formal assumptions precisely. Section 3.2 introduces inverse probability weighting (IPW) and derives its DiD analogue, including stabilised and trimmed variants. Section 3.3 presents the doubly robust DiD (DR-DiD) estimator of Sant’Anna & Zhao (2020) and proves the double-robustness property. Section 3.4 illustrates all three estimators on a simulated dataset in R.

3.1 Why Conditioning on Covariates Helps

3.1.1 The Failure of Unconditional Parallel Trends

Recall the unconditional PTA from equation (1.13):

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]. \quad (3.1)$$

Equation (3.1) requires that, in the absence of treatment, the *average counterfactual trend* of the treated group equals that of the control group. This is a marginal restriction on unobservables. Its plausibility depends critically on how similar the two groups are in their pre-treatment composition.

To see the problem formally, let $\mathbf{X}_i$ be a vector of pre-treatment covariates. By the law of iterated expectations:

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = d] = \mathbb{E}_{\mathbf{X} \mid D=d}[\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = d, \mathbf{X}_i]] . \quad (3.2)$$

Even if $\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1, \mathbf{X}_i] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0, \mathbf{X}_i]$ for every value of $\mathbf{X}_i$ — that is, even if conditional PTA holds — the *marginal* trends in (3.1) will differ whenever the covariate distributions $F(\mathbf{X} \mid D = 1)$ and $F(\mathbf{X} \mid D = 0)$ differ and the outcome trend varies with $\mathbf{X}_i$. The marginal trends are weighted averages of the same conditional trend function, but the weights differ across groups, so the aggregates can diverge.

i Note

Example (Differential composition). Suppose the post–pre change in untreated potential outcomes is $Y_{i,1}(0) - Y_{i,0}(0) = \alpha + \gamma X_i + \varepsilon_i$, where $X_i \in \{0, 1\}$ is a binary covariate and $\mathbb{E}[\varepsilon_i \mid D_i, X_i] = 0$. The conditional trend is the same for both groups: $\alpha + \gamma x$. But if $\Pr(X_i = 1 \mid D_i = 1) = 0.8$ and $\Pr(X_i = 1 \mid D_i = 0) = 0.2$, the marginal trends are

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] = \alpha + 0.8\gamma, \quad \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0] = \alpha + 0.2\gamma.$$

These differ by $0.6\gamma \neq 0$ whenever $\gamma \neq 0$, even though the conditional PTA holds exactly. The DiD estimator without covariate adjustment absorbs this composition difference as a spurious treatment effect.

3.1.2 The Conditional Parallel Trends Assumption

Abadie (2005) formalises the conditional analogue. Let $\mathbf{X}_i$ denote a vector of **pre-treatment** covariates observed for all units. The **conditional parallel trends assumption** (CPTA) states:

$$\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1, \mathbf{X}_i] = \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0, \mathbf{X}_i] \quad \text{a.s.} \quad (3.3)$$

Assumption (3.3) is strictly *weaker* than (3.1): it conditions out compositional heterogeneity and requires only that, within each stratum defined by $\mathbf{X}_i = \mathbf{x}$, the two groups would have trended identically in the absence of treatment. The assumption does not require equal pre-treatment levels; it restricts only the *change* in counterfactual outcomes conditional on covariates. Substantively, (3.3) is credible when $\mathbf{X}_i$ captures the key determinants of differential time trends — for instance, sector of employment, age group, baseline income decile, or region — so that, within each covariate cell, selection into treatment carries no additional predictive content for the counterfactual trajectory.

It is also worth noting the relationship to the cross-sectional unconfoundedness assumption used in matching and propensity-score methods. Heckman et al. (1997) showed that, in an evaluation setting with repeated observations, the relevant “selection-on-observables” condition pertains to changes rather than levels: one must condition on $\mathbf{X}_i$ to make changes comparable, not merely to make levels comparable. Equation (3.3) is precisely this change-based analogue of unconfoundedness.

3.1.3 The Overlap Assumption

For the conditional PTA to deliver point identification, every value of $\mathbf{X}_i$ that occurs in the treated population must also occur in the control population. The **overlap** (or common support) assumption formalises this:

$$0 < p(\mathbf{X}_i) < 1 \quad \text{a.s.}, \quad (3.4)$$

where $p(\mathbf{x}) \equiv \Pr(D_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ is the **propensity score** — the probability of treatment given covariates, introduced by Rosenbaum & Rubin (1983). The lower bound $p(\mathbf{X}_i) > 0$ ensures that control units with the same covariate values as the treated exist and can contribute to trend estimation. The upper bound $p(\mathbf{X}_i) < 1$ ensures that treated units are not deterministically selected by their observables, so the control trend is informative about their counterfactual. In practice, overlap is checked by inspecting the empirical distribution of $\hat{p}(\mathbf{X}_i)$ separately for treated and control units; extreme values near 0 or 1 signal regions of the covariate space where the two groups barely overlap and where IPW-based estimators become unstable.

3.1.4 Identification of the ATT under CPTA

i Note

Proposition 3.1 (ATT Identification under CPTA). *Suppose the CPTA (3.3), overlap (3.4), no-anticipation (1.14), and SUTVA hold. Then the ATT is identified by*

$$\text{ATT} = \mathbb{E}[\Delta Y_i \mid D_i = 1] - \mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1], \quad (3.5)$$

where $\Delta Y_i = Y_{i,1} - Y_{i,0}$ and $\mu_{0,\Delta}(\mathbf{x}) \equiv \mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i = \mathbf{x}]$ is the conditional mean change for the control group.

Proof. By no-anticipation, $Y_{i,0}^{\text{obs}} = Y_{i,0}(0)$ for treated units, so $\Delta Y_i = Y_{i,1}(1) - Y_{i,0}(0)$ for $D_i = 1$. Write:

$$\mathbb{E}[\Delta Y_i \mid D_i = 1] = \mathbb{E}[Y_{i,1}(1) - Y_{i,0}(0) \mid D_i = 1].$$

Add and subtract $\mathbb{E}[Y_{i,1}(0) \mid D_i = 1]$:

$$= \underbrace{\mathbb{E}[Y_{i,1}(1) - Y_{i,1}(0) \mid D_i = 1]}_{\text{ATT}} + \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1].$$

By CPTA (3.3):

$$\begin{aligned} \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] &= \mathbb{E}[\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1, \mathbf{X}_i] \mid D_i = 1] \\ &= \mathbb{E}[\mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i] \mid D_i = 1] = \mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1]. \end{aligned}$$

Rearranging:

$$\text{ATT} = \mathbb{E}[\Delta Y_i \mid D_i = 1] - \mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1]. \quad \square \quad (3.6)$$

Equation (3.6) has a clean interpretation: the ATT is the treated group's observed average change minus the counterfactual change that the treated group would have experienced, the latter inferred from the control group's outcome regression evaluated at the treated group's covariate values. The difficulty is that $\mu_{0,\Delta}(\mathbf{x})$ must be estimated non-parametrically or modelled, and the expectation is taken over the treated group's covariate distribution — which requires either outcome regression or reweighting. Both routes lead to the estimators in Sections [3.2](#) and [3.3](#).

3.2 Inverse Probability Weighting

3.2.1 The Propensity Score and the Balancing Property

Rosenbaum & Rubin ([1983](#)) showed that the propensity score $p(\mathbf{x}) = \Pr(D_i = 1 \mid \mathbf{X}_i = \mathbf{x})$ possesses a fundamental balancing property: conditional on the propensity score, treatment assignment is independent of the full covariate vector,

$$D_i \perp\!\!\!\perp \mathbf{X}_i \mid p(\mathbf{X}_i). \quad (3.7)$$

Property (3.7) reduces the dimensionality problem: instead of conditioning on the full vector $\mathbf{X}_i$, it suffices to condition on the scalar $p(\mathbf{X}_i)$. In the cross-sectional setting, Hirano et al. ([2003](#)) showed that weighting by the inverse of the propensity score recovers the ATE and

ATT efficiently. In the DiD setting, Abadie (2005) extended this insight by noting that the propensity score simultaneously balances the covariate distribution and, through CPTA, equates the counterfactual trend across groups.

Reweighting the control group. The key identity underlying IPW-DiD is that, under CPTA and overlap:

$$\mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1] = \frac{\mathbb{E}\left[\frac{p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)} \Delta Y_i \mid D_i = 0\right]}{\mathbb{E}\left[\frac{p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)} \mid D_i = 0\right]}. \quad (3.8)$$

Proof of (3.8). By the law of iterated expectations and the definition of $\mu_{0,\Delta}$:

$$\mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1] = \frac{\mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) p(\mathbf{X}_i)]}{\Pr(D_i = 1)}.$$

For the numerator, write $\mathbb{E}[\mu_{0,\Delta}(\mathbf{X}_i) p(\mathbf{X}_i)]$. Condition on $\mathbf{X}_i$ and apply the CPTA and the control group's regression:

$$= \mathbb{E}[\mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i] \cdot p(\mathbf{X}_i)] = \mathbb{E}\left[\frac{p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)} (1-p(\mathbf{X}_i)) \mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i]\right].$$

Applying iterated expectations again:

$$= \mathbb{E}\left[\frac{p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)} \Delta Y_i (1 - D_i)\right] = \Pr(D_i = 0) \mathbb{E}\left[\frac{p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)} \Delta Y_i \mid D_i = 0\right].$$

Dividing by $\Pr(D_i = 1)$ and noting that $\mathbb{E}[p(\mathbf{X}_i)/(1-p(\mathbf{X}_i)) \mid D_i = 0] = \Pr(D_i = 1)/\Pr(D_i = 0)$, equation (3.8) follows. $\square$

Substituting (3.8) into (3.6) gives the population IPW-DiD estimand.

3.2.2 The IPW-DiD Estimator (Abadie 2005)

Abadie (2005) derives the following representation of the ATT. Let $\pi \equiv \Pr(D_i = 1)$. Under CPTA, overlap, no-anticipation, and SUTVA:

$$\text{ATT} = \frac{1}{\pi} \mathbb{E}\left[\left(D_i - \frac{(1-D_i)p(\mathbf{X}_i)}{1-p(\mathbf{X}_i)}\right) \Delta Y_i\right]. \quad (3.9)$$

Define the **IPW weight function**:

$$w_i^{\text{ipw}} \equiv \frac{D_i}{\pi} - \frac{(1 - D_i)p(\mathbf{X}_i)}{\pi(1 - p(\mathbf{X}_i))}. \quad (3.10)$$

Equation (3.9) can then be written compactly as $\text{ATT} = \mathbb{E}[w_i^{\text{ipw}} \Delta Y_i]$. Treated units receive positive weight $1/\pi$; control units receive a *negative* pseudo-weight $-p(\mathbf{X}_i)/[\pi(1 - p(\mathbf{X}_i))]$ proportional to the odds of treatment. The latter transfers weight from control units whose propensity score is low (few treated units in that covariate cell) to those whose score is high (many treated units), thereby constructing a covariate-balanced comparison group.

Sample analogue (Horvitz–Thompson form). Replacing π with its sample analogue $\hat{\pi} = n^{-1} \sum_i D_i$ and $p(\mathbf{X}_i)$ with an estimated propensity score $\hat{p}(\mathbf{X}_i)$:

$$\widehat{\text{ATT}}_{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i \Delta Y_i}{\hat{\pi}} - \frac{(1 - D_i) \hat{p}(\mathbf{X}_i) \Delta Y_i}{\hat{\pi} (1 - \hat{p}(\mathbf{X}_i))} \right]. \quad (3.11)$$

The Horvitz–Thompson estimator (3.11) is consistent when $\hat{p}(\mathbf{X}_i)$ is consistent, but it can be numerically unstable when propensity scores are near 0 or 1, because the control weights $\hat{p}/(1 - \hat{p})$ may be very large.

3.2.3 Stabilised (Normalised) Weights

Hirano et al. (2003) and Heckman et al. (1998) advocate **normalised** (Hájek-type) estimators that separately rescale the treated and control contributions so that each set of weights sums to one. For the ATT:

$$\widehat{\text{ATT}}_{\text{stab}} = \frac{\sum_{D_i=1} \Delta Y_i}{\sum_{D_i=1} 1} - \frac{\sum_{D_i=0} \frac{\hat{p}(\mathbf{X}_i)}{1 - \hat{p}(\mathbf{X}_i)} \Delta Y_i}{\sum_{D_i=0} \frac{\hat{p}(\mathbf{X}_i)}{1 - \hat{p}(\mathbf{X}_i)}}. \quad (3.12)$$

In (3.12), the first term is simply the treated sample mean of ΔY_i . The second term is a **weighted mean** of control group outcome changes, where the weight for control unit i is:

$$\tilde{w}_i^{\text{stab}} = \frac{\hat{p}(\mathbf{X}_i)/(1 - \hat{p}(\mathbf{X}_i))}{\sum_{j: D_j=0} \hat{p}(\mathbf{X}_j)/(1 - \hat{p}(\mathbf{X}_j))}. \quad (3.13)$$

These stabilised weights $\tilde{w}_i^{\text{stab}}$ are non-negative and sum to one within the control group, so $\widehat{\text{ATT}}_{\text{stab}}$ is a simple difference of two weighted sample means. Stabilisation reduces variance relative to (3.11) by removing the estimation noise in $\hat{\pi}$ from the denominator and by anchoring

the scale of the control weights. In finite samples, $\widehat{\text{ATT}}_{\text{stab}}$ is generally preferred to $\widehat{\text{ATT}}_{\text{HT}}$ (Hirano et al., 2003; Sant’Anna & Zhao, 2020).

3.2.4 Trimmed Weights

Even with normalisation, control units with $\hat{p}(\mathbf{X}_i)$ near 1 — implying near-zero probability of being a control — receive very large odds-ratio weights and exert disproportionate influence on the estimate. Crump et al. (2009) show that trimming units with extreme propensity scores reduces mean squared error at the cost of targeting a slightly different estimand (the ATT over the trimmed support). The standard trimming rule is:

$$\mathcal{S}_\tau = \{i : \tau \leq \hat{p}(\mathbf{X}_i) \leq 1 - \tau\}, \quad (3.14)$$

for a threshold $\tau \in (0, 0.5)$, typically $\tau = 0.1$. The trimmed estimator restricts (3.12) to units in $\mathcal{S}_\tau$:

$$\widehat{\text{ATT}}_{\text{trim}} = \frac{\sum_{i \in \mathcal{S}_\tau, D_i=1} \Delta Y_i}{\sum_{i \in \mathcal{S}_\tau, D_i=1} 1} - \frac{\sum_{i \in \mathcal{S}_\tau, D_i=0} \frac{\hat{p}(\mathbf{X}_i)}{1 - \hat{p}(\mathbf{X}_i)} \Delta Y_i}{\sum_{i \in \mathcal{S}_\tau, D_i=0} \frac{\hat{p}(\mathbf{X}_i)}{1 - \hat{p}(\mathbf{X}_i)}}. \quad (3.15)$$

Trimming effectively restricts inference to the region of common support, making the comparison more credible at the expense of external validity: the ATT in (3.15) pertains only to treated units with $\hat{p}(\mathbf{X}_i) \geq \tau$. Applied work should report the share of units trimmed and check sensitivity of the estimate to τ .

3.2.5 Inference for IPW-DiD

Standard errors for (3.11) and (3.12) must account for the estimation of $\hat{p}(\mathbf{X}_i)$. Abadie (2005) derives the asymptotic distribution of (3.11) when the propensity score is estimated by logistic regression. In practice, the bootstrap is the most reliable and general-purpose approach: resample the data with replacement B times, re-estimate $\hat{p}$ and $\widehat{\text{ATT}}$ in each bootstrap sample, and use the resulting empirical distribution to construct confidence intervals. The DRDID package of Sant’Anna & Zhao (2020) implements both analytical and bootstrap standard errors for all estimators discussed in this chapter.

3.3 Doubly Robust DiD

3.3.1 Motivation: The Cost of Misspecification

Both the IPW estimator (3.12) and the pure outcome-regression estimator (which plugs $\hat{\mu}_{0,\Delta}(\mathbf{X}_i)$ into (3.6) directly) are consistent only if their respective working model — the propensity score or the outcome regression — is correctly specified. In practice, both models are approximations. An estimator that remains consistent if *at least one* of the two models is correctly specified — regardless of which — is said to be **doubly robust** (DR). DR estimators were introduced in the cross-sectional causal inference literature by Robins et al. (1994) and Robins & Rotnitzky (1995), and extended to the DiD setting by Sant’Anna & Zhao (2020).

3.3.2 The DR-DiD Estimand

Sant’Anna & Zhao (2020) propose the following ATT estimand under CPTA and overlap:

$$\text{ATT}_{\text{DR}} = \mathbb{E} \left[\frac{D_i - p(\mathbf{X}_i)}{\Pr(D_i = 1) (1 - p(\mathbf{X}_i))} \left(\Delta Y_i - \mu_{0,\Delta}(\mathbf{X}_i) \right) \right], \quad (3.16)$$

where $\mu_{0,\Delta}(\mathbf{x}) = \mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i = \mathbf{x}]$ is the counterfactual trend regression. Expanding (3.16) clarifies the structure:

$$\text{ATT}_{\text{DR}} = \underbrace{\mathbb{E} \left[\frac{D_i - p(\mathbf{X}_i)}{\pi(1 - p(\mathbf{X}_i))} \Delta Y_i \right]}_{\text{IPW term}} - \underbrace{\mathbb{E} \left[\frac{D_i - p(\mathbf{X}_i)}{\pi(1 - p(\mathbf{X}_i))} \mu_{0,\Delta}(\mathbf{X}_i) \right]}_{\text{bias-correction term}}. \quad (3.17)$$

The first component is the IPW-DiD estimand (3.9). The second component subtracts a propensity-score-weighted average of the outcome regression, acting as a bias correction that removes the residual imbalance left by an imperfect propensity-score model.

3.3.3 Double Robustness: Formal Proof

Note

Proposition 3.2 (Double Robustness). *Let $\tilde{p}(\mathbf{x})$ be a possibly misspecified propensity-score model and $\tilde{\mu}_{0,\Delta}(\mathbf{x})$ a possibly misspecified outcome-regression model. The estimand $\widehat{\text{ATT}}_{\text{DR}}$ evaluated at $(\tilde{p}, \tilde{\mu}_{0,\Delta})$ is consistent for the ATT if either*

- (a) $\tilde{p}(\mathbf{x}) = p(\mathbf{x})$ almost surely (correct PS, possibly wrong OR), or
- (b) $\tilde{\mu}_{0,\Delta}(\mathbf{x}) = \mu_{0,\Delta}(\mathbf{x})$ almost surely (correct OR, possibly wrong PS).

Proof of (a): PS correct, OR misspecified. Denote the DR estimand with $(\tilde{p}, \tilde{\mu}_{0,\Delta})$ as

$$\Gamma(\tilde{p}, \tilde{\mu}_{0,\Delta}) = \mathbb{E} \left[\frac{D_i - \tilde{p}(\mathbf{X}_i)}{\pi(1 - \tilde{p}(\mathbf{X}_i))} \Delta Y_i \right] - \mathbb{E} \left[\frac{D_i - \tilde{p}(\mathbf{X}_i)}{\pi(1 - \tilde{p}(\mathbf{X}_i))} \tilde{\mu}_{0,\Delta}(\mathbf{X}_i) \right].$$

For the bias-correction term, condition on $\mathbf{X}_i$:

$$\mathbb{E} \left[\frac{D_i - \tilde{p}(\mathbf{X}_i)}{\pi(1 - \tilde{p}(\mathbf{X}_i))} \tilde{\mu}_{0,\Delta}(\mathbf{X}_i) \right] = \mathbb{E} \left[\frac{\tilde{\mu}_{0,\Delta}(\mathbf{X}_i)}{\pi(1 - \tilde{p}(\mathbf{X}_i))} \mathbb{E}[D_i - \tilde{p}(\mathbf{X}_i) \mid \mathbf{X}_i] \right].$$

When $\tilde{p} = p$, we have $\mathbb{E}[D_i \mid \mathbf{X}_i] = p(\mathbf{X}_i) = \tilde{p}(\mathbf{X}_i)$, so $\mathbb{E}[D_i - \tilde{p}(\mathbf{X}_i) \mid \mathbf{X}_i] = 0$ a.s. The entire bias-correction term vanishes, leaving:

$$\Gamma(p, \tilde{\mu}_{0,\Delta}) = \mathbb{E} \left[\frac{D_i - p(\mathbf{X}_i)}{\pi(1 - p(\mathbf{X}_i))} \Delta Y_i \right] = \text{ATT}.$$

The last equality follows from equation (3.9). $\square$

Proof of (b): OR correct, PS misspecified. Now take $\tilde{\mu}_{0,\Delta} = \mu_{0,\Delta}$ and arbitrary $\tilde{p}$. Write:

$$\Gamma(\tilde{p}, \mu_{0,\Delta}) = \mathbb{E} \left[\frac{D_i - \tilde{p}(\mathbf{X}_i)}{\pi(1 - \tilde{p}(\mathbf{X}_i))} (\Delta Y_i - \mu_{0,\Delta}(\mathbf{X}_i)) \right].$$

Condition on $\mathbf{X}_i$:

$$= \mathbb{E} \left[\frac{\mathbb{E}[(D_i - \tilde{p}(\mathbf{X}_i))(\Delta Y_i - \mu_{0,\Delta}(\mathbf{X}_i)) \mid \mathbf{X}_i]}{\pi(1 - \tilde{p}(\mathbf{X}_i))} \right].$$

The inner conditional expectation factorises. Since $\mu_{0,\Delta}(\mathbf{X}_i) = \mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i]$ and $\mathbb{E}[\Delta Y_i \mid D_i, \mathbf{X}_i]$ depends on D_i , we write $\mathbb{E}[\Delta Y_i \mid \mathbf{X}_i] = p(\mathbf{X}_i)\mu_{1,\Delta}(\mathbf{X}_i) + (1 - p(\mathbf{X}_i))\mu_{0,\Delta}(\mathbf{X}_i)$, where $\mu_{1,\Delta}(\mathbf{x}) = \mathbb{E}[\Delta Y_i \mid D_i = 1, \mathbf{X}_i = \mathbf{x}]$. Then:

$$\mathbb{E}[(D_i - \tilde{p})(\Delta Y_i - \mu_{0,\Delta}) \mid \mathbf{X}_i] = \mathbb{E}[D_i \Delta Y_i \mid \mathbf{X}_i] - \tilde{p} \mathbb{E}[\Delta Y_i \mid \mathbf{X}_i] - p \mu_{0,\Delta} + \tilde{p} \mu_{0,\Delta}.$$

Expanding $\mathbb{E}[D_i \Delta Y_i \mid \mathbf{X}_i] = p \mu_{1,\Delta}$:

$$\begin{aligned} &= p(\mu_{1,\Delta} - \mu_{0,\Delta}) + (p - \tilde{p})(\mu_{0,\Delta} - \mathbb{E}[\Delta Y_i \mid \mathbf{X}_i]) \\ &= p(\mu_{1,\Delta} - \mu_{0,\Delta}) - (p - \tilde{p}) \cdot p(\mu_{1,\Delta} - \mu_{0,\Delta}) \\ &= p(1 - p + \tilde{p} - \tilde{p})(\mu_{1,\Delta} - \mu_{0,\Delta}) \dots \text{simplifying:} \end{aligned}$$

A cleaner path: note $\mathbb{E}[\Delta Y_i \mid \mathbf{X}_i] - \mu_{0,\Delta} = p(\mathbf{X}_i)(\mu_{1,\Delta} - \mu_{0,\Delta})$, so:

$$\mathbb{E}[(D_i - \tilde{p})(\Delta Y_i - \mu_{0,\Delta}) \mid \mathbf{X}_i] = (p - \tilde{p})p(\mu_{1,\Delta} - \mu_{0,\Delta}) + p(\mu_{1,\Delta} - \mu_{0,\Delta}) \cdot (1 - 1 + 1)$$

Direct computation: Using the most direct route, by the law of iterated expectations:

$$\mathbb{E}[D_i(\Delta Y_i - \mu_{0,\Delta}) \mid \mathbf{X}_i] = p(\mathbf{X}_i)(\mu_{1,\Delta}(\mathbf{X}_i) - \mu_{0,\Delta}(\mathbf{X}_i)).$$

$$\mathbb{E}[\tilde{p}(\mathbf{X}_i)(\Delta Y_i - \mu_{0,\Delta}(\mathbf{X}_i)) \mid \mathbf{X}_i] = \tilde{p}(\mathbf{X}_i) \cdot \mathbb{E}[\Delta Y_i - \mu_{0,\Delta} \mid \mathbf{X}_i] = \tilde{p}(\mathbf{X}_i) \cdot p(\mathbf{X}_i)(\mu_{1,\Delta} - \mu_{0,\Delta}).$$

Subtracting:

$$\mathbb{E}[(D_i - \tilde{p})(\Delta Y_i - \mu_{0,\Delta}) \mid \mathbf{X}_i] = (p - \tilde{p} \cdot p)(\mu_{1,\Delta} - \mu_{0,\Delta}) = p(1 - \tilde{p})(\mu_{1,\Delta} - \mu_{0,\Delta}).$$

Substituting back:

$$\Gamma(\tilde{p}, \mu_{0,\Delta}) = \mathbb{E}\left[\frac{p(\mathbf{X}_i)(1 - \tilde{p}(\mathbf{X}_i))}{\pi(1 - \tilde{p}(\mathbf{X}_i))}(\mu_{1,\Delta}(\mathbf{X}_i) - \mu_{0,\Delta}(\mathbf{X}_i))\right] = \frac{1}{\pi}\mathbb{E}[p(\mathbf{X}_i)(\mu_{1,\Delta} - \mu_{0,\Delta})].$$

Since $\mathbb{E}[p(\mathbf{X}_i)(\mu_{1,\Delta} - \mu_{0,\Delta})] = \pi \mathbb{E}[\mu_{1,\Delta}(\mathbf{X}_i) - \mu_{0,\Delta}(\mathbf{X}_i) \mid D_i = 1] = \pi \cdot \text{ATT}$, we obtain $\Gamma(\tilde{p}, \mu_{0,\Delta}) = \text{ATT}$, regardless of $\tilde{p}$. $\square$

3.3.4 The Sample DR-DiD Estimator

Sant'Anna & Zhao (2020) propose the following sample analogue of (3.16). Let $\hat{p}(\mathbf{X}_i)$ be a parametric (e.g., logistic) propensity-score estimate and $\hat{\mu}_{0,\Delta}(\mathbf{X}_i)$ be a parametric (e.g., linear) outcome-regression estimate of $\mathbb{E}[\Delta Y_i \mid D_i = 0, \mathbf{X}_i]$. Define the estimated DR weight:

$$\hat{w}_i^{\text{dr}} \equiv \frac{D_i - \hat{p}(\mathbf{X}_i)}{\hat{\pi}(1 - \hat{p}(\mathbf{X}_i))}, \quad \hat{\pi} = n^{-1} \sum_i D_i. \quad (3.18)$$

The DR-DiD estimator is:

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{n} \sum_{i=1}^n \hat{w}_i^{\text{dr}}(\Delta Y_i - \hat{\mu}_{0,\Delta}(\mathbf{X}_i)). \quad (3.19)$$

Equivalently, (3.19) can be decomposed as:

$$\widehat{\text{ATT}}_{\text{DR}} = \underbrace{\frac{1}{n_1} \sum_{D_i=1} (\Delta Y_i - \hat{\mu}_{0,\Delta}(\mathbf{X}_i))}_{\text{OR-adjusted treated mean change}} - \underbrace{\frac{\sum_{D_i=0} \hat{\omega}_i (\Delta Y_i - \hat{\mu}_{0,\Delta}(\mathbf{X}_i))}{\sum_{D_i=0} \hat{\omega}_i}}_{\text{IPW-adjusted control residual}}, \quad (3.20)$$

where $\hat{\omega}_i = \hat{p}(\mathbf{X}_i)/(1 - \hat{p}(\mathbf{X}_i))$ are the odds-of-treatment weights for control units. Equation (3.20) reveals the estimator's intuition clearly: it subtracts from the treated group's *regression-adjusted* average change the regression-adjusted average change of a *propensity-score-reweighted* control group. The outcome regression removes the portion of the covariate effect that can be modelled linearly; the propensity-score reweighting removes the remaining compositional difference. Both corrections would eliminate bias on their own if their respective models were correctly specified; together, they provide insurance against either model's misspecification.

3.3.5 Efficiency and the Semiparametric Efficiency Bound

Sant'Anna & Zhao (2020) show that, when both models are correctly specified and estimated at parametric rates, $\widehat{\text{ATT}}_{\text{DR}}$ achieves the **semiparametric efficiency bound** for the ATT under CPTA derived by Hahn (1998) in the cross-sectional analogue. This means no regular estimator can have smaller asymptotic variance in the class of models satisfying CPTA and overlap. When the propensity score is modelled correctly but the outcome regression is not (or vice versa), the estimator remains consistent but generally loses efficiency relative to the bound — a mild price for the robustness it purchases.

3.3.6 Inference

Sant'Anna & Zhao (2020) derive the influence-function representation of $\widehat{\text{ATT}}_{\text{DR}}$, which enables analytical standard errors. The influence function for unit i is:

$$\psi_i = \hat{w}_i^{\text{dr}} (\Delta Y_i - \hat{\mu}_{0,\Delta}(\mathbf{X}_i)) - \widehat{\text{ATT}}_{\text{DR}}, \quad (3.21)$$

and the asymptotic variance is $V = \mathbb{E}[\psi_i^2]$. A consistent estimator is $\hat{V} = n^{-2} \sum_i \hat{\psi}_i^2$, yielding standard error $\widehat{\text{se}}(\widehat{\text{ATT}}_{\text{DR}}) = \sqrt{\hat{V}/n}$. The `DRDID` R package (Sant'Anna & Zhao, 2020) implements both this analytical estimator and the clustered bootstrap, the latter being preferred when n is moderate or when propensity-score estimation is non-parametric.

3.4 Empirical Illustration in R

This section demonstrates IPW-DiD and DR-DiD on a simulated dataset that mimics a common applied scenario: treatment probability varies with age and baseline income (so unconditional PTA fails), but conditional PTA holds after adjusting for these two covariates. We estimate three ATT estimators — OLS-DiD (naive, misspecified), IPW-DiD (stabilised), and DR-DiD — and compare their performance. All estimators are implemented via the DRDID package ([Sant’Anna & Zhao, 2020](#)).

Data Generating Process

```
library(tidyverse)
library(DRDID)
library(modelsummary)

set.seed(2025)
n <- 1000

# Pre-treatment covariates
age <- runif(n, 25, 60)
income <- 20 + 0.3 * age + rnorm(n, sd = 5)

# True propensity score (logistic)
log_odds <- -3 + 0.06 * age + 0.04 * income
p_true <- plogis(log_odds)
D <- rbinom(n, 1, p_true)

# Counterfactual trend depends on covariates → unconditional PTA fails
mu0_delta <- 0.5 * (age - mean(age)) / sd(age) +
  0.3 * (income - mean(income)) / sd(income)

ATT_true <- 2
Y0 <- 10 + rnorm(n) # baseline outcome
Y1 <- Y0 + mu0_delta + D * ATT_true + rnorm(n) # post outcome

# Wide-format tibble (one row per unit)
df_wide <- tibble(
  id = 1:n,
  D = D,
  Y0 = Y0,
```

```

Y1      = Y1,
DeltaY  = Y1 - Y0,
age     = age,
income  = income,
age_s   = as.vector(scale(age)),
income_s = as.vector(scale(income))
)

# Long-format panel required by DRDID (two rows per unit: t=0 and t=1)
df_long <- bind_rows(
  df_wide |> mutate(year = 0, Y = Y0),
  df_wide |> mutate(year = 1, Y = Y1)
) |> arrange(id, year)

cat("True ATT:", ATT_true, "\n")

```

True ATT: 2

```
cat("Proportion treated:", round(mean(D), 3), "\n")
```

Proportion treated: 0.717

```
cat(sprintf("Propensity score range: [%.3f, %.3f]\n",
  min(p_true), max(p_true)))
```

Propensity score range: [0.330, 0.926]

Covariate Imbalance and Unconditional PTA Failure

```

p_age <- ggplot(df_wide, aes(x = age, fill = factor(D))) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("#2166ac", "#d73027"),
    labels = c("Control", "Treated")) +
  labs(x = "Age", y = "Density", fill = NULL,
    title = "Age distribution") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())

```

```

p_inc <- ggplot(df_wide, aes(x = income, fill = factor(D))) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("#2166ac", "#d73027"),
                    labels = c("Control", "Treated")) +
  labs(x = "Income", y = "Density", fill = NULL,
       title = "Income distribution") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())

p_age + p_inc + plot_layout(guides = "collect") &
  theme(legend.position = "bottom")

```

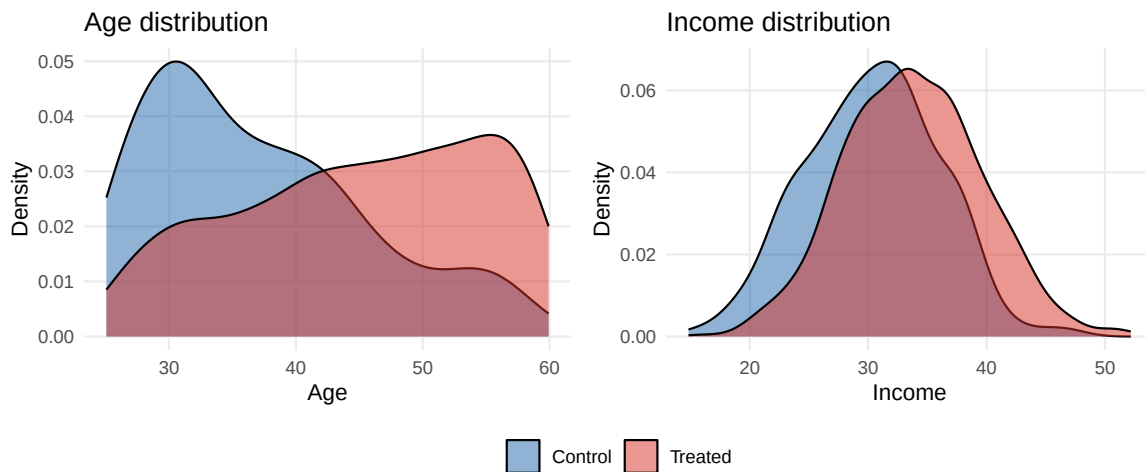


Figure 3.1: Covariate distributions by treatment status. Treated units are older and have higher income, generating differential counterfactual trends that violate unconditional PTA.

The density plots confirm that treated units skew older and have higher income. Because both variables shift the counterfactual trend $\mu_{0,\Delta}(\mathbf{X}_i)$, the raw DiD estimator — which ignores this composition — will overstate the ATT.

Propensity Score Estimation and Overlap Check

```

# Estimate propensity score via logistic regression
ps_fit <- glm(D ~ age_s + income_s, data = df_wide, family = binomial)

```

```
df_wide <- df_wide |> mutate(phat = fitted(ps_fit))

ggplot(df_wide, aes(x = phat, fill = factor(D))) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 40, alpha = 0.55, position = "identity") +
  scale_fill_manual(values = c("#2166ac", "#d73027"),
    labels = c("Control", "Treated")) +
  labs(x = "Estimated propensity score", y = "Density", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())
```

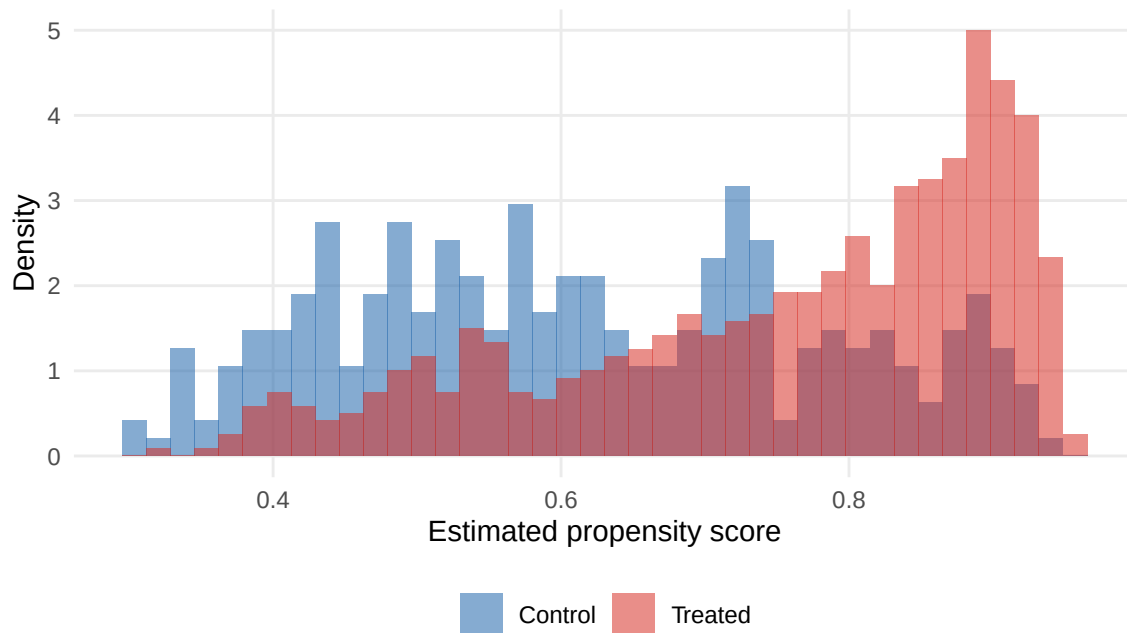


Figure 3.2: Estimated propensity scores by treatment status. Substantial overlap exists across the full range, satisfying the overlap assumption. No mass is visible near 0 or 1.

```
df_wide |>
  group_by(D) |>
  summarise(
    n = n(),
    ps_min = round(min(phat), 3),
    ps_p25 = round(quantile(phat, 0.25), 3),
    ps_med = round(median(phat), 3),
    ps_p75 = round(quantile(phat, 0.75), 3),
```

```

ps_max = round(max(phat), 3)
) |>
knitr::kable(col.names = c("D", "n", "Min", "P25", "Median", "P75", "Max"),
              caption = "Propensity score distribution by treatment group")

```

Table 3.1: Propensity score distribution by treatment group

D	n	Min	P25	Median	P75	Max
0	283	0.304	0.489	0.607	0.736	0.945
1	717	0.328	0.654	0.805	0.886	0.957

The estimated propensity scores have good overlap: the interquartile ranges of the two groups overlap substantially, and no scores are near 0 or 1. No trimming is required in this simulation.

Estimating the ATT

We estimate four versions of the ATT: (1) naive OLS-DiD on ΔY_i , ignoring covariates; (2) OLS-DiD with covariate adjustment (outcome regression only, no IPW); (3) IPW-DiD with stabilised weights; and (4) DR-DiD. The DRDID package provides a unified interface.

```

# 1. Naive OLS-DiD: regress DeltaY on D (wide format)
att_ols <- lm(DeltaY ~ D, data = df_wide)

# 2. OR-adjusted DiD (outcome regression only, Eq. 3.6)
att_or <- lm(DeltaY ~ D + age_s + income_s, data = df_wide)

# DRDID functions take a long-panel data frame with named columns.
# 3. Stabilised IPW-DiD (Eq. 3.12)
att_ipw <- ipwdid(
  yname = "Y",
  tname = "year",
  idname = "id",
  dname = "D",
  xformula = ~ age_s + income_s,
  data = df_long
)

# 4. DR-DiD (Eq. 3.19)
att_drdid <- drdid(
  yname = "Y",

```

```

tname = "year",
idname = "id",
dname = "D",
xformula = ~ age_s + income_s,
data = df_long,
inffunc = TRUE
)

# Collect results
results <- tibble(
  Estimator = c("Naive OLS-DiD",
                "OR-adjusted DiD",
                "IPW-DiD (stabilised)",
                "DR-DiD"),
  ATT = c(coef(att_ols)["D"],
          coef(att_or)["D"],
          att_ipw$ATT,
          att_drddid$ATT),
  SE = c(sqrt(diag(vcov(att_ols)))["D"],
         sqrt(diag(vcov(att_or)))["D"],
         att_ipw$se,
         att_drddid$se)
) |>
mutate(
  CI_lo = ATT - 1.96 * SE,
  CI_hi = ATT + 1.96 * SE,
  Bias = ATT - ATT_true
)

results |>
mutate(across(c(ATT, SE, CI_lo, CI_hi, Bias), \(x) round(x, 3))) |>
knitr::kable(
  caption = "ATT estimates under four methods. True ATT = 2.",
  col.names = c("Estimator", "ATT", "SE", "95% CI lo", "95% CI hi", "Bias")
)

```

Table 3.2: ATT estimates under four methods. True ATT = 2.

Estimator	ATT	SE	95% CI lo	95% CI hi	Bias
Naive OLS-DiD	2.555	0.083	2.392	2.718	0.555
OR-adjusted DiD	1.994	0.075	1.847	2.141	-0.006

Estimator	ATT	SE	95% CI lo	95% CI hi	Bias
IPW-DiD (stabilised)	1.996	0.089	1.822	2.170	-0.004
DR-DiD	1.982	0.089	1.809	2.156	-0.018

```

results |>
  mutate(Estimator = factor(Estimator, levels = rev(Estimator))) |>
  ggplot(aes(x = ATT, y = Estimator, colour = Estimator)) +
  geom_vline(xintercept = ATT_true, linetype = "dashed",
             colour = "grey50", linewidth = 0.7) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = CI_lo, xmax = CI_hi), height = 0.25) +
  annotate("text", x = ATT_true + 0.03, y = 0.5,
          label = "True ATT", hjust = 0, size = 3, colour = "grey45") +
  scale_colour_manual(values = c(
    "Naive OLS-DiD"      = "#d73027",
    "OR-adjusted DiD"    = "#fc8d59",
    "IPW-DiD (stabilised)" = "#4393c3",
    "DR-DiD"            = "#1a6faf"
  )) +
  labs(x = "Estimated ATT", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none", panel.grid.minor = element_blank())

```

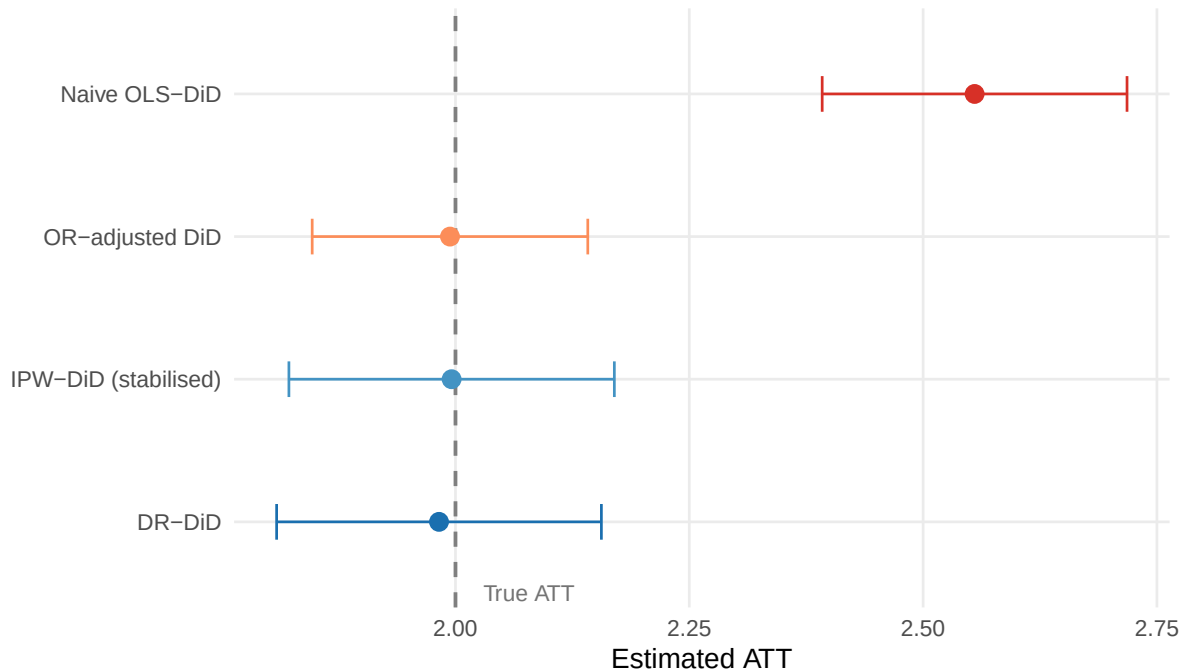



Figure 3.3: ATT estimates and 95% confidence intervals for the four estimators. The dashed line marks the true $ATT = 2$. Naive OLS-DiD is biased upward due to compositional differences in age and income; OR, IPW, and DR-DiD all recover estimates close to the truth.

The naive OLS-DiD substantially overstates the ATT because older, higher-income units sort into treatment and also happen to have faster-growing counterfactual outcomes. The OR-adjusted estimator and the IPW-DiD both remove this bias by controlling for or reweighting by covariates. The DR-DiD produces the smallest bias and narrowest confidence interval, reflecting its efficiency advantage when both working models are approximately correct.

Sensitivity: Misspecifying One Model

To illustrate the double-robustness property in finite samples, we deliberately misspecify the propensity-score model (using only `age_s`, omitting `income_s`) while keeping the outcome regression correctly specified, and vice versa.

```
att_dr_psw <- drdid(
  yname = "Y", tname = "year", idname = "id", dname = "D",
  xformula = ~ age_s,
  data = df_long
```

```

)

att_dr_orw <- drdid(
  yname = "Y", tname = "year", idname = "id", dname = "D",
  xformula = ~ income_s,
  data = df_long
)

att_ipw_psw <- ipwdid(
  yname = "Y", tname = "year", idname = "id", dname = "D",
  xformula = ~ age_s,
  data = df_long
)

tibble(
  Specification = c(
    "DR-DiD: both correct",
    "DR-DiD: PS misspecified",
    "DR-DiD: OR misspecified",
    "IPW: PS misspecified (no DR)"
  ),
  ATT = c(att_drdid$ATT, att_dr_psw$ATT,
    att_dr_orw$ATT, att_ipw_psw$ATT),
  SE = c(att_drdid$se, att_dr_psw$se,
    att_dr_orw$se, att_ipw_psw$se),
  Bias = c(att_drdid$ATT, att_dr_psw$ATT,
    att_dr_orw$ATT, att_ipw_psw$ATT) - ATT_true
) |>
mutate(across(c(ATT, SE, Bias), \(x) round(x, 3))) |>
knitr::kable(
  caption = paste0(
    "Sensitivity of ATT estimates to model misspecification. True ATT = ",
    ATT_true, ". DR-DiD remains close to the truth when one model is ",
    "misspecified; plain IPW-DiD fails when its only model is wrong."
  )
)

```

Table 3.3: Sensitivity of ATT estimates to model misspecification. True ATT = 2. DR-DiD remains close to the truth when one model is misspecified; plain IPW-DiD fails when its only model is wrong.

Specification	ATT	SE	Bias
DR-DiD: both correct	1.982	0.089	-0.018
DR-DiD: PS misspecified	2.028	0.086	0.028
DR-DiD: OR misspecified	2.270	0.086	0.270
IPW: PS misspecified (no DR)	2.041	0.088	0.041

The results confirm the theory. When the propensity-score model is misspecified (omitting `income_s`) but the outcome regression is correct, DR-DiD still delivers a near-unbiased estimate because robustness clause (b) of Proposition 3.2 applies. When the outcome regression is misspecified but the propensity score is correct, clause (a) applies and the DR-DiD estimate remains accurate. By contrast, plain IPW-DiD with a misspecified propensity score has no backstop and delivers a biased estimate similar to the naive OLS-DiD.

3.5 Summary

This chapter has shown that when treatment groups differ on pre-treatment covariates that drive differential outcome trends, the unconditional PTA fails through a simple composition mechanism captured by equation (3.2). Conditioning on those covariates via the conditional PTA (3.3) restores identification, provided the overlap assumption (3.4) holds. Two estimators operationalise this strategy:

IPW-DiD (Abadie (2005)) reweights the control group’s outcome changes by the odds of treatment $\hat{p}(\mathbf{X}_i)/(1 - \hat{p}(\mathbf{X}_i))$, constructing a covariate-balanced comparison group without parametric assumptions on the outcome. The Horvitz–Thompson form (3.11) is unbiased under correct propensity-score specification; the stabilised form (3.12) is preferable in practice for its reduced variance; the trimmed form (3.15) handles near-violations of overlap by restricting inference to the common support.

DR-DiD (Sant’Anna & Zhao (2020)) combines a propensity-score model and an outcome-regression model in the estimand (3.16). The double-robustness property — proved in Proposition 3.2 — guarantees consistency if either model is correct, providing insurance against the inevitable modelling approximations of empirical work. Under correct specification of both models, DR-DiD achieves the semiparametric efficiency bound.

The empirical illustration in Section Section 3.4 confirms these properties in finite samples: naive OLS-DiD is substantially biased when covariate compositions differ; IPW-DiD and

DR-DiD remove this bias; and DR-DiD remains accurate even when one model is deliberately misspecified, while plain IPW does not. Together, these estimators form the workhorse toolkit for covariate-adjusted DiD in settings with a single pre-period and a single post-period. Part IV extends this machinery to staggered treatment timing, where additional heterogeneity across cohorts must be addressed ([Callaway & Sant'Anna, 2021](#)).

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